

Fluid Mechanics

Project Report

Title: Engineering Analysis of Long-Distance Natural Gas Transportation Through the TAPI Pipeline



Class: BEME-F-24-B

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Introduction:

Abstract:

The TAPI gas pipeline is a project that will carry natural gas from Turkmenistan to Afghanistan, Pakistan and India. This pipeline will help these countries have a supply of energy and work together more closely on trade. The pipeline is very long. Goes through different types of land which can make it hard to keep the gas flowing. When gas flows through a pipe its pressure goes down because of friction and changes in elevation. Understanding these pressure changes is crucial for ensuring that the gas flows smoothly and reaches its destination with pressure. This project looks at how the pressure of the gas changes as it flows through the pipeline and how the elevation of the land effects this. We also study how compressor stations help keep the gas pressure up. The TAPI gas pipeline and gas transportation are the focus of this project. We want to know how well the pipeline works and if it needs any help to deliver gas to its destination.

Description:

The TAPI Gas Pipeline is a natural gas pipeline that runs about 1817 km from the Galkynysh Gas Field in Turkmenistan to Fazilka at the India–Pakistan border. The pipeline passes through elevations and this causes pressure losses due to pipe friction and changes in terrain height. The goal of this project is to use mechanics to calculate pressure changes and study how the pipeline works. By looking at these factors, we can determine how efficiently the pipeline transports gas and if it needs compressor stations to maintain the required pressure.

Aim of Project:

The aim of this project is to study the TAPI natural gas pipeline focusing on pressure drop Hydraulic Grade Line (HGL) and Energy Grade Line (EGL) along the pipeline. We want to understand how friction and elevation changes affect long-distance gas transportation.

OBJECTIVES:

Primary Objectives:

- The TAPI gas pipeline and how it works are the focus.
- We study the concept of pressure drop in long-distance gas pipelines.
- We look at Hydraulic Grade Line (HGL) and Energy Grade Line (EGL).
- We analyze how friction and elevation changes affect gas flow.

Secondary Objectives:

- We study the role of compressor stations, in maintaining gas pressure.
- We examine the mechanics used in pipeline systems.

- The engineering significance of the TAPI gas pipeline is also considered.

THEORETICAL FOUNDATION:

1. Basic Terminology of Gas Pipelines

Continuity Principle

- Based on the conservation of mass, meaning gas flow remains continuous in the pipeline.
- The amount of gas entering a section equals the amount leaving (assuming no leakage).
- Helps maintain steady gas transportation in the TAPI pipeline.

Bernoulli's Principle

- Explains the conservation of energy in fluid flow.
- Pressure, velocity, and elevation energy change throughout the pipeline.
- Helps understand pressure variation due to friction and terrain elevation.

Darcy–Weisbach Principle

- Used to explain pressure loss due to pipe wall friction.
- Pressure drop increases with pipeline length and gas velocity.
- Important for analyzing long pipelines like TAPI.

Reynolds Number Concept

- Determines whether flow is laminar or turbulent.
- TAPI pipeline mostly experiences turbulent flow because of high gas velocity.
- Turbulent flow increases frictional losses.

Head Loss Principle

- Energy loss occurs because of friction and pipe fittings.
- Includes **major losses** (pipe friction) and **minor losses** (valves, elbows, tees).
- Affects pressure throughout the pipeline.

HGL and EGL Concept

- HGL shows pressure and elevation energy changes.

- EGL represents total fluid energy in the pipeline.
- Used to analyze energy loss and pressure distribution.

2. Working Principle of Natural Gas Pipelines:

- Natural gas flows through the pipeline due to a **pressure difference**, moving from high pressure to low pressure.
- As gas travels, **friction and elevation changes** cause pressure loss along the pipeline.
- **Compressor stations** are installed to restore pressure and maintain continuous gas flow over long distances like the TAPI pipeline.

Literature Review:

Several studies explain that fluid mechanics principles are important in long-distance pipeline systems for understanding pressure drop, energy loss, and flow behavior. Research shows that friction, pipe roughness, velocity, and elevation changes significantly affect pressure variation in pipelines. In the TAPI gas pipeline, theoretical analysis becomes important because the pipeline extends approximately 1817 km across varying terrain elevations, where pressure losses and energy changes can affect gas transportation efficiency. Concepts such as Hydraulic Grade Line (HGL) and Energy Grade Line (EGL) are commonly used to study energy distribution and pipeline performance.

Previous Researches

Several researchers have studied long-distance gas transmission pipelines to understand pressure loss, flow behavior, and energy changes during transportation. Studies explain that frictional resistance, pipe roughness, flow velocity, and terrain elevation significantly affect pressure variation in pipelines. Researchers also emphasize the importance of hydraulic analysis in maintaining efficient gas transportation over large distances. Furthermore, concepts such as Hydraulic Grade Line (HGL) and Energy Grade Line (EGL) are commonly used to analyze energy distribution and pressure changes in pipeline systems.

Comparison from Literature

From the reviewed literature, it is observed that different studies use various methods to analyze pipeline performance depending on pipeline length, operating pressure, and terrain conditions. However, most studies agree that long-distance pipelines experience considerable pressure loss due to friction and elevation changes. In the case of the TAPI gas pipeline, which extends approximately 1817 km across varying terrain, theoretical fluid mechanics analysis becomes essential to understand pressure behavior and ensure efficient gas delivery.

Calculations:

Given Data:

Total Length of Pipe = 1817 km

Fluid = Natural Gas (Methane Gas)

Flow Rate $Q = 33$ billion cubic meters = $1046.4 \text{ m}^3/\text{s}$

Diameter $D = 56 \text{ in} = 1.4224 \text{ m}$

Pipe = high-grade steel pipeline

$\varepsilon = 0.000046 \text{ m}$

KL Factor Values:

Sr. No	Shapes / Bends	KL Value
1	30° Threaded Elbow	0.25
2	45° Threaded Elbow	3.75
3	Tee Flanged through branch	1.0
4	Fully Open Ball Valve	0.05

Nomenclature:

Symbol	Description	Unit
P	Pressure	Pa
V	Fluid Velocity	m/s
D	Diameter	m
L	Length	m
F	Friction factor	-
G	gravity	m/s^2
H_L	head loss	m
HGL	Hydraulic Grade Line	-
EGL	Energy Grade Line	-
Re	Reynolds Number	-
Q	Flow rate	m^3/s
ρ	Density	kg / m^3
μ	Dynamic Viscosity	Pa.s
γ	Specific Weight	N / m^3
ε	Pipe roughness	m

Fluid Properties:

Cities names	Country	Temperature	Altitude (m)	Density	D. Viscosity (micro Pa.s)	K. viscosity	s. weight	Pipes	major losses	Bends	minor losses	(hL*s.w)/major	(hL*s.w) minor	(s.w*z) m		
Galkynysh	Turkmenistan	2	225	ρ_1	0.694	10.18	14.6685879	Y_1	6.80814	10km	1530310.666	Ball Valve	1104.2506	10418569.25	7517.89268	1531.8315
Mary		2	223	ρ_2	0.701			10.57	15.07845934	Y_2	6.87681	40km	6127041.581			42134500.82
Murgap		3.1	231	ρ_3	0.699	10.6	15.16452074	Y_3	6.85719	40km	6127279.113			42015917.06	0	1584.01089
Yoloten		1	259	ρ_4	0.7	10.58	15.11428571	Y_4	6.867	40km	6127140.469			42075073.6	0	1778.553
Tagtabazar		3.3	342	ρ_5	0.701	10.58	15.09272468	Y_5	6.87681	47km	7199320.125			49508356.63	0	2351.86902
Serhetabat		4	747	ρ_6	0.675	10.24	15.17037037	Y_6	6.62175	15km	2297735.721	El 2	8840.7623	15215031.51	58541.31776	4946.44725
			747	ρ_6	0.675			Y_6	6.62175		Ball Valve	1104.2506	0	7312.071411	4946.44725	
			747	ρ_6	0.675			Y_6	6.62175		El 3	8840.7623	0	58541.31776	4946.44725	
Torghundi		1	650	ρ_7	0.714	10.3	14.42577031	Y_7	7.00434	66km	10106643.78			70790369.31	0	4552.821
Islam Qala		0	680	ρ_8	0.717	10.3	14.36541144	Y_8	7.03377	66km	10106368.46			71085871.29	0	4782.9636
Injil	3	920	ρ_9	0.706	10.4	14.73087819	Y_9	6.92586	68km	10414339.03			72128254.12	0	6371.7912	
Afghanistan	Herat	3	920	ρ_{10}	0.706	10.4	14.73087819	Y_{10}	6.92586	20km	2594053.183	El 4	8840.7623	17966049.18	61229.88198	6371.7912
			920	ρ_{10}	0.706			Y_{10}	6.92586		Tee	22101.9057	0	153074.7046	6371.7912	
			920	ρ_{10}	0.706			Y_{10}	6.92586		Ball Valve (branch)	7.0777	0	49.01915932	6371.7912	
			920	ρ_{10}	0.706			Y_{10}	6.92586		El 5	7482.8212	0	51824.97204	6371.7912	
	Pashtun Zarghun	1	1042	ρ_{11}	0.714	10.3	14.42577031	Y_{11}	7.00434	50km	6484166.092			45417303.93	0	7298.52228
	Adraskan	-1	1343	ρ_{12}	0.722	10.25	14.1966759	Y_{12}	7.08282	50km	6483439.641			45921035.96	0	9512.22726
	Shindand	4	1066	ρ_{13}	0.71	10.35	14.57746479	Y_{13}	6.9651	50km	6484646.89			45166214.05	0	7424.7966
	Anar Darah	6	720	ρ_{14}	0.7005	10.34	14.76088508	Y_{14}	6.871905	50km	6485228.008			44565870.77	0	4947.7716
	Farah	7	650	ρ_{15}	0.698	10.38	14.87106017	Y_{15}	6.84738	22km	1915351.47	El 6	7482.8212	13115139.35	51237.72023	4450.797
			650	ρ_{15}	0.698			Y_{15}	6.84738		Ball Valve	935.3526	0	6404.714686	4450.797	
			650	ρ_{15}	0.698			Y_{15}	6.84738		El 7	3127.5959	0	21415.83761	4450.797	
	Delaram	6	817	ρ_{16}	0.7005	10.34	14.76088508	Y_{16}	6.871905	63km	5473378.087			37612534.24	0	5614.346385
	Girishk	6	817	ρ_{17}	0.7005	10.34	14.76088508	Y_{17}	6.871905	63km	5473378.087	Valve	625.5192	37612534.24	4298.508518	5614.346385
	Lashkargah	8	762	ρ_{18}	0.6955	10.42	14.98202732	Y_{18}	6.82855	62km	5387203.564			36756108.77	0	5199.01551
	Kandahar	7.5	1016	ρ_{19}	0.6967	10.4	14.92751543	Y_{19}	6.834627	15km + 15km	2113410.042			14444369.34	0	6943.981032
			1016	ρ_{19}	0.6967			Y_{19}	6.834627	5km	240214.76	El 8	3127.5959	1641778.284	21375.95138	6943.981032
			1016	ρ_{19}	0.6967			Y_{19}	6.834627			Tee	12510.3837	0	85503.80622	6943.981032
			1016	ρ_{19}	0.6967			Y_{19}	6.834627			Ball Valve (branch)	6.255	0	42.75059189	6943.981032
			1016	ρ_{19}	0.6967			Y_{19}	6.834627			Ball Valve	506.6648	0	3462.864922	6943.981032
			1016	ρ_{19}	0.6967			Y_{19}	6.834627			El 9	2751.0811	0	18802.61317	6943.981032
	Spin Boldak	4.85	1222	ρ_{20}	0.703	10.5	14.93598862	Y_{20}	6.89643	65km	3114377.574			21478086.94	0	8427.43746
	Wesh	3.85	1290	ρ_{21}	0.706	10.46	14.81586402	Y_{21}	6.92586	65km	3114084.532			21567713.49	0	8934.3594
Chaman	6.4	1329	ρ_{22}	0.6995	10.36	14.81057898	Y_{22}	6.862095	50km	2029541.157	El 10	2751.0811	13926904.23	18878.17986	9119.724255	
		1329	ρ_{22}	0.6995			Y_{22}	6.862095			Tee	6877.6	0	47194.74457	9119.724255	
		1329	ρ_{22}	0.6995			Y_{22}	6.862095			Ball Valve (branch)	2.2	0	15.096609	9119.724255	
		1329	ρ_{22}	0.6995			Y_{22}	6.862095			Ball Valve	291.0644	0	1997.311564	9119.724255	
		1329	ρ_{22}	0.6995			Y_{22}	6.862095			El 11	2328.5151	0	15978.49183	9119.724255	
Mastung	2	1800	ρ_{23}	0.7107	10.17	14.30983537	Y_{23}	6.971967	55km	2231547.808			15558277.68	0	12549.5406	
Quetta	4.8	1680	ρ_{24}	0.7035	10.28	14.61265103	Y_{24}	6.901335	55km + 37km	3733729.684	El 12	2328.5151	25767719.35	16069.86276	11594.2428	
		1680	ρ_{24}	0.7035			Y_{24}	6.901335			El 13	2328.5151	0	16069.86276	11594.2428	
Sibi	Pakistan	13	130	ρ_{25}	0.6833	10.62	15.54222157	Y_{25}	6.703173	56.25km	2284645.987			15314377.3	0	871.41249
Dera Bugti		13.5	110	ρ_{26}	0.6821	10.64	15.59888579	Y_{26}	6.691401	56.25km	2284755.179			15288213.09	0	736.05411
Rajanpur		13	112	ρ_{27}	0.6833	10.62	15.54222157	Y_{27}	6.703173	56.25km	2284645.987			15314377.3	0	750.755376
Dera ghazi Khan		12.5	147	ρ_{28}	0.6845	10.6	15.48575603	Y_{28}	6.714945	56.25km	2284537.153			15340541.33	0	987.096915
Taunsa Sharif		12	190	ρ_{29}	0.6858	10.58	15.42723826	Y_{29}	6.727698	56.25km	2284424.336			15368917.04	0	1278.26262
Muzaffargarh		13	123	ρ_{30}	0.6833	10.62	15.54222157	Y_{30}	6.703173	56.25km	2284645.987			15314377.3	0	824.490279
Kot Addu		13	113	ρ_{31}	0.6833	10.62	15.54222157	Y_{31}	6.703173	56.25km	2284645.987			15314377.3	0	757.458549
Multan		13.7	168	ρ_{32}	0.6816	10.65	15.625	Y_{32}	6.686496	56.25km	2284805.493	El 14	2328.5151	15277342.79	15569.6069	1123.331328
		13.7	168	ρ_{32}	0.6816			Y_{32}	6.686496			Tee	5821.2877	0	38924.01692	1123.331328
		13.7	168	ρ_{32}	0.6816			Y_{32}	6.686496			Ball Valve (branch)	35.65	0	238.3735824	1123.331328
	13.7	168	ρ_{32}	0.6816	Y_{32}			6.686496			El 15	983.7976	0	6578.158717	1123.331328	
Lodhran	India	13	130	ρ_{33}	0.6833	10.62	15.54222157	Y_{33}	6.703173	61km	1054038.863			7065404.845	0	871.41249
Vehari		13	135	ρ_{34}	0.6833	10.62	15.54222157	Y_{34}	6.703173	61km	1054038.863			7065404.845	0	904.928355
Fazilka		10	182	ρ_{35}	0.69	10.7	15.50724638	Y_{35}	6.7689	61km	1053992.931			7134372.75	0	1231.9398

Formulas Used:

Equation	Formula	Purpose
Darcy-Weisbach	$h_f = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$	Major losses
Haaland Equation	$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\frac{\varepsilon/D}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right]$	Friction factor
Minor Loss	$h_m = K \cdot \frac{V^2}{2g}$	Elbows, tees, valves
Energy Equation	$P_1 = P_2 + \rho g(z_2 - z_1) + \rho g \cdot h_L$	Backward pressure calc
HGL	$HGL = \frac{P}{\rho g} + z$	Hydraulic Grade Line
EGL	$EGL = \frac{P}{\rho g} + z + \frac{V^2}{2g}$	Energy Grade Line

Graphs:

Elevation Pressure:

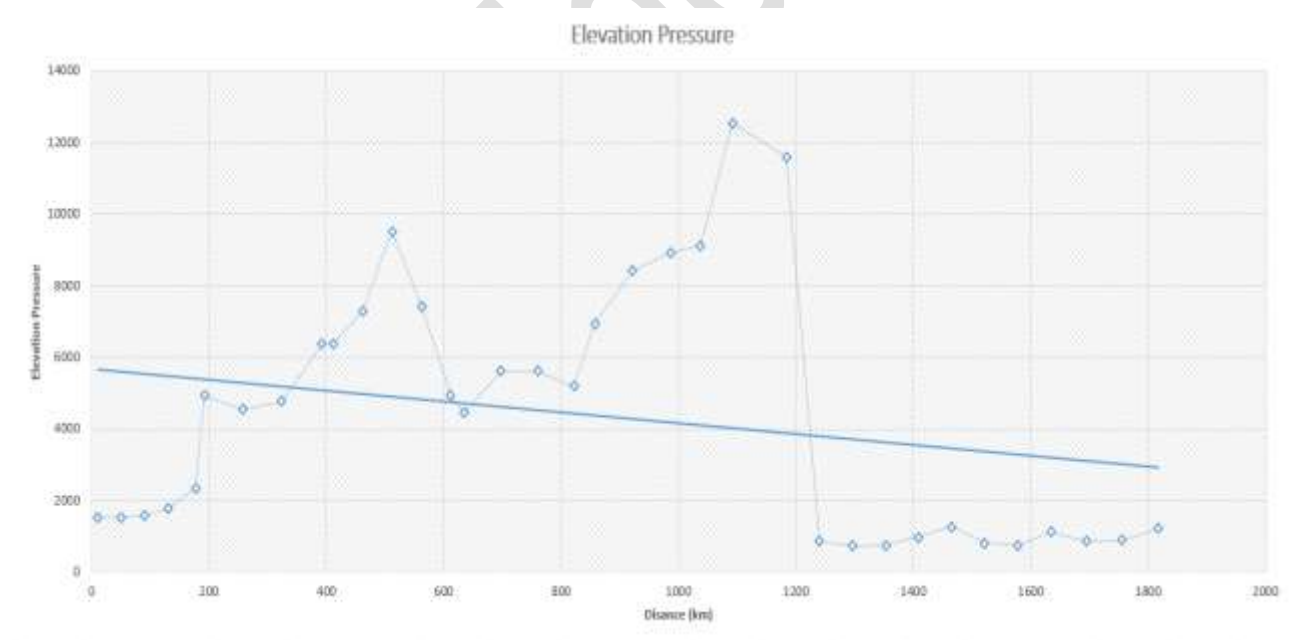


Figure 1

Pressure Drop:

Distance	Pressure Drop
192	201562965.3
412	232221349.2

612	181126146.2
852	139662874.8
987	44772396
1184	55305491.3
1817	143825127

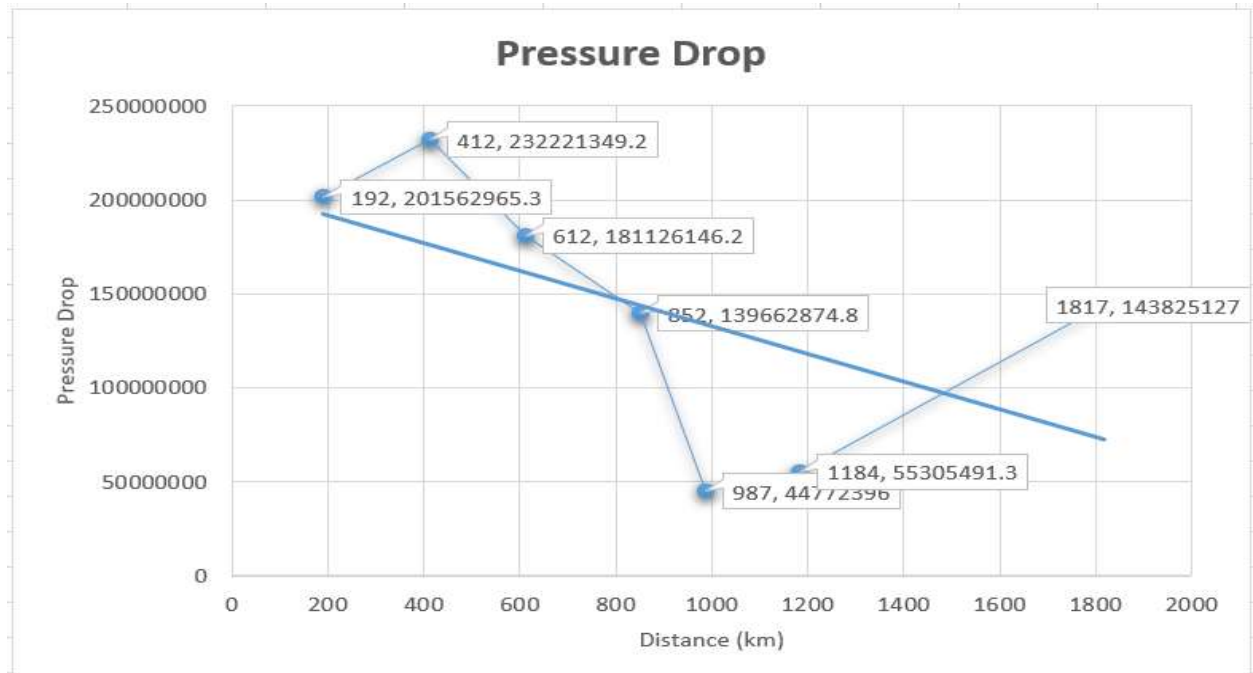


Figure 2

Pressure at Different Points:

Distance	Pressure
0	10132500
192	191430465.3
412	423651814.5
612	604777960.7
612	362866776.4
852	223203901.6
852	580330144.2
987	535557748.2
1184	480252256.9
1817	336427129.9

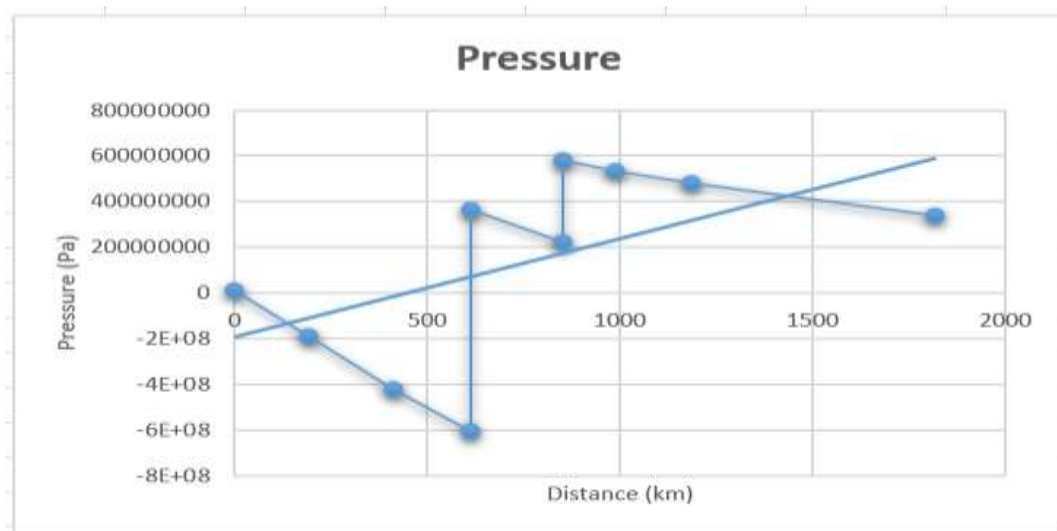


Figure 3

HGL:

Distance	HGL
0	1488516.956
192	-28908599.5
412	-61168640.8
612	-88321885.1
612	52994171.08
852	32658818.92
852	84911303.6
987	78047136.91
1184	69589992.54
1817	49702072.99

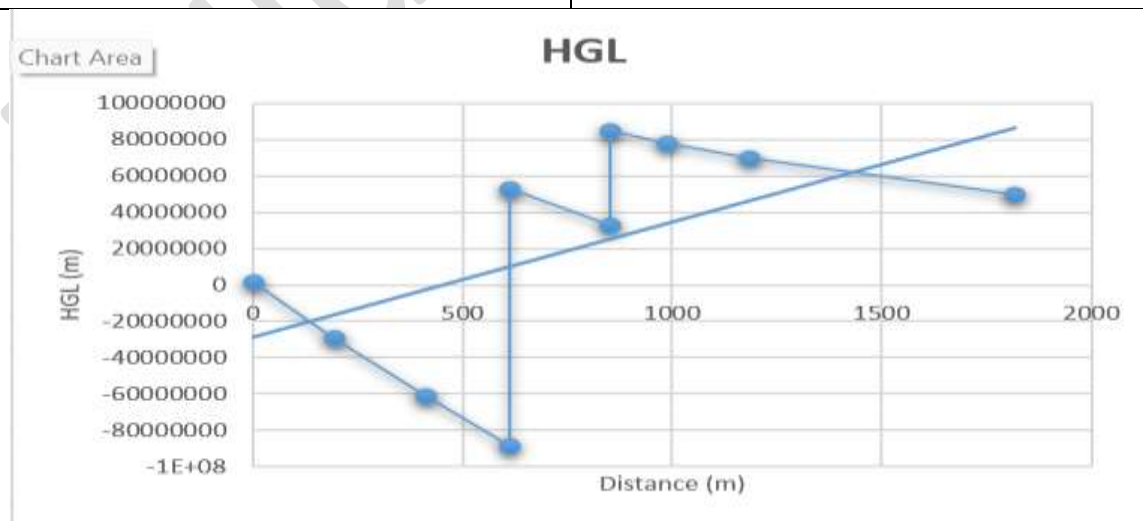


Figure 4

EGL:

Distance	EGL
0	1510617.989
192	8886498.49
412	61149935.74
612	88303180.07
612	53006681.32
852	32668952.35
852	84918181.21
987	78052958.19
1184	69595813.82
1817	49704532.48

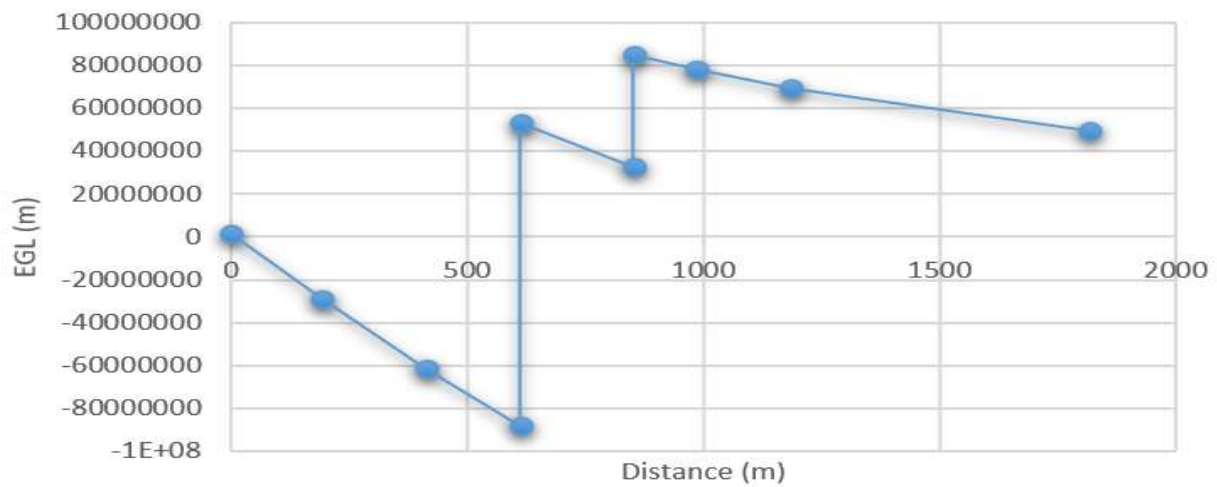
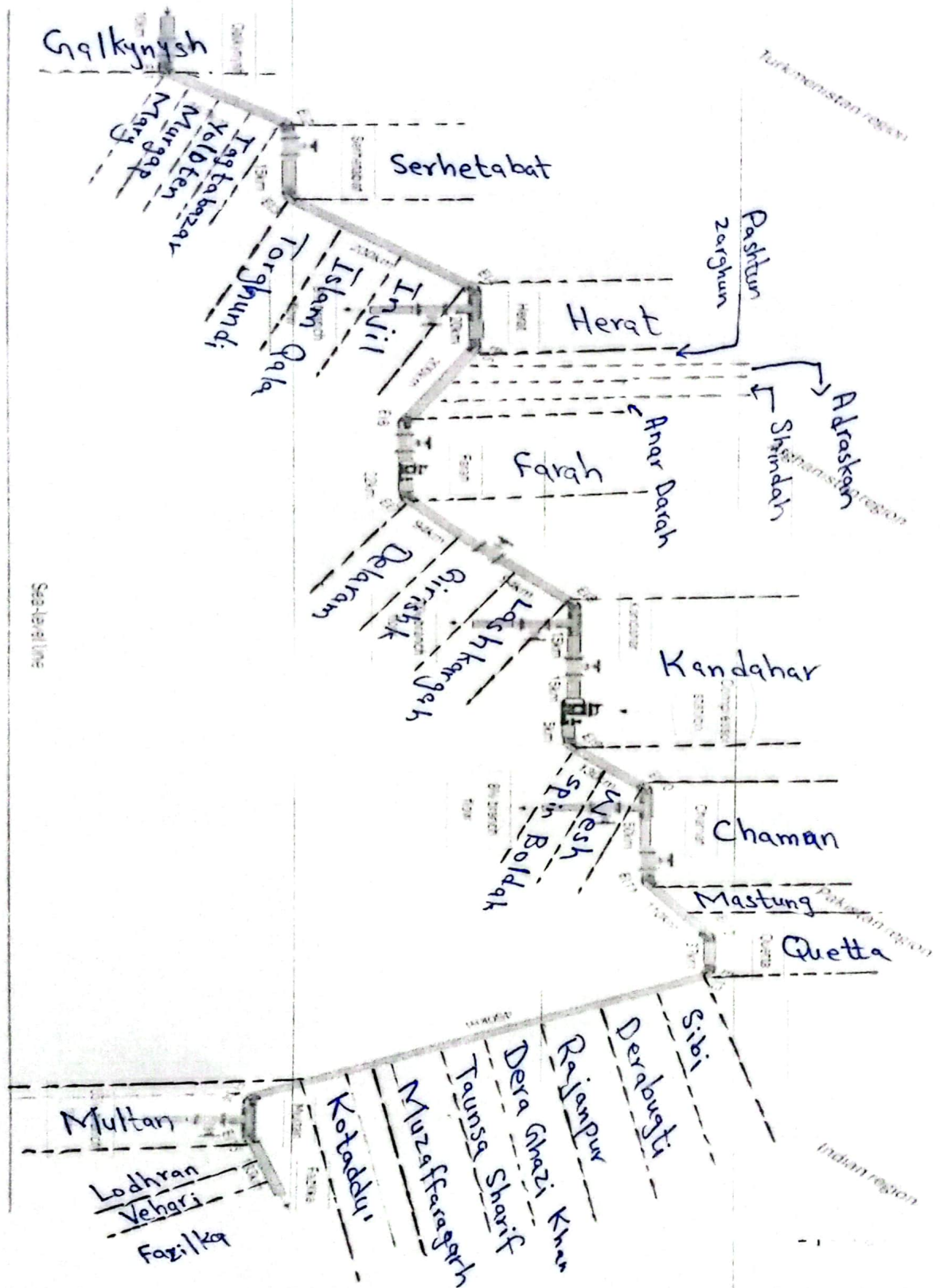
EGL

Figure 5

Hand-Written Calculations:

Handwritten Calculations are given Below



HEAD LOSSES

Galkynysh:

$$\rightarrow Re = \frac{\rho v d}{\mu}$$

$$Re = \frac{0.694 \times 658.5 \times 1.4224}{10.18 \times 10^{-6}}$$

$$Re = 6.38 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon/D}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.38 \times 10^7} \right] \right]^2}$$

$$f = 0.009856$$

For 10 km pipe:

$$h_{L, \text{major}} = f \frac{L}{D} \frac{v^2}{2g}$$

$$h_{L, \text{major}} = 0.009856 \times \frac{10000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L, \text{major}} = 1530310.6656 \text{ m}$$

For Ball valve:

$$h_{L, \text{minor}} = K_L \frac{v^2}{2g}$$

$$h_{L, \text{minor}} = 0.05 \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L, \text{minor}} = 1104.2506 \text{ m}$$

For EL 1:

$$h_{L \text{ minor}} = K_L \frac{V_1^2}{2g}$$

$$h_{L \text{ minor}} = 0.4 \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 8840.7623 \text{ m}$$

167km pipe (In 4 parts):

Mary (40km):

$$\rightarrow Re = \frac{\rho_2 V_1 d}{\mu_2}$$

$$Re = \frac{0.701 \times 658.5 \times 1.4224}{10.57 \times 10^{-6}}$$

$$Re = 6.21 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{8.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.21 \times 10^7} \right] \right]^2}$$

$$f = 0.009858$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009858 \times \frac{40000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 6127041.5812 \text{ m}$$

Murgap (40km):

$$\rightarrow Re = \frac{\rho_3 V_1 d}{\mu_3}$$

$$Re = \frac{0.699 \times 658.5 \times 1.4224}{10.6 \times 10^{-6}}$$

$$Re = 6.18 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.18 \times 10^7} \right] \right]^2}$$

$$f = 0.009858$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009858 \times \frac{40000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 6127279.1134 \text{ m}$$

Yoloten (40km):

$$\rightarrow Re = \frac{\rho_4 V_1 d}{\mu_4}$$

$$Re = \frac{0.7 \times 658.5 \times 1.4224}{10.58 \times 10^{-6}}$$

$$Re = 6.1972 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.1972 \times 10^7} \right] \right]^2}$$

$$f = 0.009858$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009858 \times \frac{40000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 6127140.469 \text{ m}$$

Tagtabazar (47 km):

$$\rightarrow Re = \frac{\rho_s V_1 d}{\mu_s}$$

$$Re = \frac{0.701 \times 658.5 \times 1.4224}{10.58 \times 10^{-6}}$$

$$Re = 6.21 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7 D} \right)^{1.11} + \frac{6.9}{Re} \right]^2 \right]}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.21 \times 10^7} \right]^2 \right]}$$

$$f = 0.009858$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009858 \times \frac{47000}{1.4224} \times \frac{(658.5)^2}{(2 \times 9.81)}$$

$$h_{L \text{ major}} = 7199300.1247 \text{ m}$$

Serhetabat:

$$\rightarrow Re = \frac{\rho_s V_1 d}{\mu_s}$$

$$Re = \frac{0.675 \times 658.5 \times 1.4224}{10.24 \times 10^{-6}}$$

$$Re = 6.17 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{D(3.7)} \right)^{1.11} + \frac{6.9}{Re} \right]^2 \right]}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{1.4224 \times 3.7} \right)^{1.11} + \frac{6.9}{6.17 \times 10^7} \right]^2 \right]}$$

$$f = 0.009858$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

15 km pipe

$$h_{L \text{ major}} = 0.009858 \times \frac{15000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2297735.721$$

$\rightarrow$ Elbow El 2 and 3: (same velocity and K_L)

$$h_{L \text{ minor}} = (2 \times 0.4) \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 17681.5246 \text{ m}$$

$\rightarrow$ Ball Valve:

$$h_{L \text{ minor}} = K_L \frac{V_1^2}{2g}$$

$$h_{L \text{ minor}} = 0.05 \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 1104.2506 \text{ m}$$

Torglund: (66 km):

$$\rightarrow Re = \frac{\rho_f V_1 d}{\mu_f}$$

$$Re = \frac{0.714 \times 658.5 \times 1.4224}{10.3 \times 10^{-6}}$$

$$Re = 6.49 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.49 \times 10^7} \right] \right]^2}$$

$$f = 0.009855$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009855 \times \frac{66000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 10106643.78 \text{ m}$$

Islam Qala (66km):

$$\rightarrow Re = \frac{\rho_g V_1 d}{\mu_g}$$

$$Re = \frac{0.717 \times 658.5 \times 1.4224}{10.3 \times 10^{-6}}$$

$$Re = 6.52 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{D \times 3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.52 \times 10^7} \right] \right]^2}$$

$$f = 0.009855$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009855 \times \frac{66000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 10106368.46 \text{ m}$$

Injil (68km):

$$\rightarrow Re = \frac{\rho_g V_1 d}{\mu_g}$$

$$Re = \frac{0.706 \times 658.5 \times 1.4224}{10.4 \times 10^{-6}}$$

$$Re = 6.36 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{D \times 3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.36 \times 10^7} \right] \right]^2}$$

$$f = 0.009856$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_1^2}{2g}$$

$$h_{L \text{ major}} = 0.009856 \times \frac{68000}{1.4224} \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 10414339.0314 \text{ m}$$

Herat: (Before 8% branch)

→ Elbow E14

$$h_{L \text{ minor}} = K_L \frac{V_1^2}{2g}$$

$$h_{L \text{ minor}} = 0.4 \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 8840.7623 \text{ m}$$

→ Tee:

$$h_{L \text{ minor}} = K_L \frac{V_1^2}{2g}$$

$$h_{L \text{ minor}} = 1 \times \frac{(658.5)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 22101.9057$$

→ Solving velocity after branch

$$Q_2 = 1046.4 \times \frac{8}{100} = 83.712 \text{ m}^3/\text{s} \quad [8\% \text{ flow}]$$

$$Q_3 = 1046.4 - 83.712 = 962.688 \text{ m}^3/\text{s} \quad [92\% \text{ flow}]$$

$$V_2 = \frac{Q_2}{A} = \frac{83.712 \text{ m}^3/\text{s}}{\frac{\pi}{4} (1.4224)^2} = 52.7 \text{ m/s}$$

$$V_3 = \frac{Q_3}{A} = \frac{962.688 \text{ m}^3/\text{s}}{\frac{\pi}{4} (1.4224)^2} = 605.8 \text{ m/s}$$

→ Branch Ball Valve:

$$h_{L \text{ minor}} = K_L \frac{V_2^2}{2g}$$

$$h_{L \text{ minor}} = 0.05 \times \frac{(52.7)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 7.0777 \text{ m}$$

(After 8% branch)

$$\rightarrow Re = \frac{\rho V_3 d}{\mu_{10}}$$

$$Re = \frac{0.706 \times 605.8 \times 1.4224}{10.4 \times 10^{-6}}$$

$$Re = 5.85 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{5.85 \times 10^7} \right] \right]^2}$$

$$f = 0.009861$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_3^2}{2g}$$

$$h_{L \text{ major}} = 0.009861 \times \frac{20,000}{1.4224} \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2594053.1827 \text{ m}$$

$\rightarrow$ Elbow El 5:

$$h_{L \text{ minor}} = K_L \frac{V_3^2}{2g}$$

$$h_{L \text{ minor}} = 0.05 \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 7482.8212 \text{ m}$$

Pashtun Zarghun (50km):

$$\rightarrow Re = \frac{\rho V_3 d}{\mu_{11}}$$

$$Re = \frac{0.714 \times 605.8 \times 1.4224}{10.3 \times 10^{-6}}$$

$$Re = 5.97 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{5.97 \times 10^7} \right] \right]^2}$$

$$f = 0.009861$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_3^2}{2g}$$

$$h_{L \text{ major}} = 0.009861 \times \frac{50000}{1.4224} \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 6484166.092 \text{ m}$$

Adrasan (50 km):

$$\rightarrow Re = \frac{\rho V_3 d}{\mu_2}$$

$$Re = \frac{0.722 \times 605.8 \times 1.4224}{10.25 \times 10^{-6}}$$

$$Re = 6.07 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{6.07 \times 10^7} \right] \right]^2}$$

$$f = 0.009859$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_3^2}{2g}$$

$$h_{L \text{ major}} = 0.009857 \times \frac{50000}{1.4224} \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 6483439.641 \text{ m}$$

Shindand (50 km):

$$\rightarrow Re = \frac{\rho V_3 d}{\mu_3}$$

$$Re = \frac{0.71 \times 605.8 \times 1.4224}{10.35 \times 10^{-6}}$$

$$Re = 5.91 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{5.91 \times 10^7} \right] \right]^2}$$

$$f = 0.009861$$

$$\rightarrow h_{L \text{ major}} = f \frac{L}{D} \frac{V_3^2}{2g}$$

$$h_{L\text{major}} = 0.009861 \times \frac{50000}{1.4224} \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L\text{major}} = 6484646.89 \text{ m}$$

Anar Darah (50 km):

$$\rightarrow Re = \frac{\rho V_3 d}{\mu}$$

$$Re = \frac{0.7005 \times 605.8 \times 1.4224}{10.34 \times 10^{-3}}$$

$$Re = 6485228.008 \text{ m} \times 5.84 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{\epsilon}{3.7D} \right)^{11} + \frac{6.9}{Re} \right] \right]^2}$$

$$f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^2 \times \frac{6.9}{5.84 \times 10^7} \right] \right]^2}$$

$$f = 0.009862$$

$$\rightarrow h_{L\text{major}} = f \frac{L}{D} \frac{V_3^2}{2g}$$

$$h_{L\text{major}} = 0.009862 \times \frac{50000}{1.4224} \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L\text{major}} = 6485228.008 \text{ m}$$

Farah: (Before Compressor)

→ Elbow El 6:

$$h_{L\text{minor}} = K_L \frac{V_3^2}{2g}$$

$$h_{L\text{minor}} = 0.4 \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L\text{minor}} = 7482.8212 \text{ m}$$

→ Ball Valve:

$$h_{L\text{minor}} = K_L \frac{V_3^2}{2g}$$

$$h_{L\text{minor}} = 0.05 \times \frac{(605.8)^2}{2 \times 9.81}$$

$$h_{L\text{minor}} = 935.3526 \text{ m}$$

→ Solving Compressor:

$$\text{Compressor inlet Pressure} = \frac{604777960.7}{4058075.2} \text{ Pa}$$

$$\text{Compression ratio} = 1.6$$

$$r = \frac{P_{\text{outlet}}}{P_{\text{inlet}}} \Rightarrow P_{\text{outlet}} = r \times P_{\text{inlet}}$$

$$P_{\text{outlet}} = 1.6 \times 604777960.7$$

$$P_{\text{outlet}} = 967644737.1$$

For Methane:

$$C_p = 2.2537 \text{ kJ/kg K}$$

$$C_v = 1.735 \text{ kJ/kg K}$$

$$k = \frac{C_p}{C_v} = \frac{2.2537}{1.735} = 1.29866$$

$$R = 0.5182 \text{ kJ/kg K} = 518.2 \text{ J/kg K}$$

$$T_1 = 280.15 \text{ K}$$

$$T_2 = T_1 r^{\left(\frac{k-1}{k}\right)}$$

$$T_2 = 280.15 \times (1.6)^{\left(\frac{1.29866-1}{1.29866}\right)}$$

$$T_2 = 311.97 \text{ K}$$

$$T_{r1} = \frac{T_1}{T_{cr}}$$

$$T_{r1} = \frac{280.15}{191.1}$$

$$T_{r1} = 1.47$$

$$Pr_1 = \frac{P_{\text{inlet}}}{P_{cr}}$$

$$Pr_1 = \frac{604.7}{4.64}$$

$$Pr_1 = 130.3$$

$$\boxed{Z = 2}$$

$$P_{\text{inlet}} = \frac{P}{ZRT} = \frac{604777960.7}{2 \times 518.2 \times 280.15}$$

$$P_{\text{inlet}} = 2082.9 \text{ kg/m}^3$$

$$T_{r2} = \frac{311.97}{191.1}$$

$$T_{r2} = 1.63$$

$$\boxed{Z = 2.35}$$

$$Pr_2 = \frac{967.6}{4.64}$$

$$Pr_2 = 208.5$$

$$\rho_{\text{outlet}} = \frac{967644737.1}{2.35 \times 518.2 \times 311.97} = 2547.05 \text{ kg/m}^3$$

$$\dot{m}_{\text{inlet}} = \dot{m}_{\text{outlet}}$$

$$(\rho Q)_{\text{inlet}} = (\rho Q)_{\text{outlet}}$$

$$Q_{\text{outlet}} = \frac{(\rho Q)_{\text{inlet}}}{\rho_{\text{outlet}}}$$

$$Q_{\text{outlet}} = \frac{2082.9 \times 962.688}{2547.05}$$

$$Q_4 = \boxed{Q_{\text{outlet}} = 787.26 \text{ m}^3/\text{s}}$$

$$V_4 = \frac{Q_{\text{outlet}}}{A} = \frac{787.26}{\pi/4 (1.4224)^2}$$

$$\boxed{V_4 = 495.43 \text{ m/s}}$$

(After Compressor)

$$\begin{aligned} \rightarrow \rho &= 0.6269 \text{ kg/m}^3 \\ \mu &= 11.49 \times 10^{-6} \text{ Pa s} \end{aligned} \quad \left. \vphantom{\begin{aligned} \rho &= 0.6269 \text{ kg/m}^3 \\ \mu &= 11.49 \times 10^{-6} \text{ Pa s} \end{aligned}} \right\} \text{ at } 311.97 \text{ K or } 38.82^\circ\text{C}$$

$$\rightarrow Re = \frac{\rho V_4 d}{\mu}$$

$$Re = \frac{0.6269 \times 495.43 \times 1.4224}{11.49 \times 10^{-6}}$$

$$Re = 3.8449 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.8449 \times 10^7} \right] \right]^2}$$

$$f = 0.009899$$

$$\rightarrow h_{L, \text{major}} = f \frac{L}{D} \frac{V_4^2}{2g}$$

$$h_{L, \text{major}} = 0.009899 \times \frac{22000}{1.4224} \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 1915351.4695 \text{ m}$$

→ Elbow El 7:

$$h_{L \text{ minor}} = 0.25 \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = \frac{5004.1535 \text{ m}}{3127.5959 \text{ m}}$$

Delaram (63km):

$$\rightarrow Re = \frac{\rho_{16} V_4 d}{\mu_{16}}$$

$$Re = \frac{0.7005 \times 495.43 \times 1.4224}{10.34 \times 10^{-6}}$$

$$Re = 4.7741 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left(\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{4.7741 \times 10^7} \right) \right]^2}$$

$$f = 0.009878$$

$$\rightarrow h_{L \text{ major}} = 0.009878 \times \frac{63000}{1.4224} \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 5473378.0867 \text{ m}$$

→

Girishk (63km):

$$Re = 4.7741 \times 10^7$$

$$f = 0.009878$$

$$h_{L \text{ major}} = 5473378.0867 \text{ m}$$

Same as Delaram because ρ , μ and Q are same as Delaram.

→ Ball Valve:

$$h_{L \text{ minor}} = 0.05 \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 625.3592 \text{ m}$$

Lashkargah (62 km):

$$\rightarrow Re = \frac{\rho_{18} V_4 d}{\mu_{18}}$$

$$Re = \frac{0.6955 \times 495.43 \times 1.4224}{10.42 \times 10^{-6}}$$

$$Re = 4.7037 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{4.7037 \times 10^7} \right] \right]}$$

$$f = 0.009879$$

$$\rightarrow h_{L \text{ major}} = 0.009879 \times \frac{62000}{1.4224} \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 5387203.5639 \text{ m}$$

Kandahar: (Before 10% branch)

$\rightarrow$ Elbow El 8:

$$h_{L \text{ minor}} = 0.25 \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 3127.5959 \text{ m}$$

$\rightarrow$ Tee:

$$h_{L \text{ minor}} = 1 \times \frac{(495.43)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 12510.3837 \text{ m}$$

Solving 10% branch:

$$Q_5 = 787.26 \times \frac{10}{100} = 78.726 \text{ m}^3/\text{s}$$

$$Q_6 = 787.26 - 78.726 = 708.534 \text{ m}^3/\text{s}$$

$$V_5 = \frac{Q_5}{A} = \frac{78.726}{\frac{\pi}{4} (1.4224)^2} = 49.54 \text{ m/s}$$

$$\underline{V_6} = \frac{Q_6}{A} = \frac{708.534}{\frac{\pi}{4} (1.4224)^2} = 445.89 \text{ m/s}$$

→ Branch Ball Valve:

$$h_{L \text{ minor}} = 0.05 \times \frac{(49.54)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 6.255 \text{ m}$$

→ Ball Valve:

$$h_{L \text{ minor}} = 0.05 \times \frac{(445.89)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 506.6648 \text{ m}$$

$$\rightarrow Re = \frac{\rho V d}{\mu}$$

$$Re = \frac{0.6967 \times 445.89 \times 1.4224}{10.4 \times 10^{-6}}$$

$$Re = 4.2487 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left(\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{11} + \frac{6.9}{4.2487 \times 10^7} \right) \right]^2}$$

$$f = 0.009889$$

$$\rightarrow h_{L \text{ major}} = 0.009889 \times \frac{30000}{1.4224} \times \frac{(445.89)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2113410.0422 \text{ m}$$

→ Solving Compressor Station:

$$P_{\text{inlet}} = 223203901.6 \text{ Pa}$$

$$T_1 =$$

$$P_{\text{outlet}} = 357126242.6 \text{ Pa}$$

$$Pr_1 = \frac{P_{\text{inlet}}}{P_{\text{cr}}}$$

$$Tr_1 = \frac{T_1}{T_{\text{cr}}}$$

$$Pr_1 = \frac{223.2}{4.64}$$

$$Tr_1 = \frac{280.65}{191.1}$$

$$Pr_1 = 48.7$$

$$Tr_1 = 1.4686$$

$$\boxed{Z = 1.42}$$

$$T_2 = T_1 r^{\left(\frac{k-1}{k}\right)}$$

$$T_2 = 280.65 \times 1.6^{\left(\frac{1.29866-1}{1.29866}\right)}$$

$$T_2 = 312.685 \text{ K}$$

$$T_{r2} = \frac{T_2}{T_{cr}}$$

$$T_{r2} = \frac{312.685}{191.1}$$

$$T_{r2} = 1.636$$

$$P_{r2} = \frac{P_{outlet}}{P_{cr}}$$

$$P_{r2} = \frac{357.1}{4.64}$$

$$P_{r2} = 76.96$$

$$\boxed{Z = 1.68}$$

$$\rho_1 = \frac{P_{in}}{Z R T_1} = \frac{223203901.6}{1.42 \times 518.2 \times 280.65} = 1080.8 \text{ kg/m}^3$$

$$\rho_2 = \frac{P_{out}}{Z R T_2} = \frac{357126242.6}{1.68 \times 518.2 \times 312.685} = 1311.9 \text{ kg/m}^3$$

$$\dot{V}_{in} Q_{in} = \dot{V}_{out} Q_{out}$$

$$Q_{out} = \frac{\dot{V}_{in} Q_{in}}{\rho_{out}}$$

$$Q_{out} = \frac{1080.8 \times 708.534}{1311.9}$$

$$Q_7 = \boxed{Q_{out} = 583.72 \text{ m}^3/\text{s}}$$

$$V_7 = \frac{Q_7}{A} = \frac{583.72}{\frac{\pi}{4} (1.4224)^2} = 367.34 \text{ m/s}$$

(After Compressor):

$$\left. \begin{aligned} \rightarrow \rho &= 0.6254 \text{ kg/m}^3 \\ \mu &= 11.513 \times 10^{-6} \text{ Pa}\cdot\text{s} \end{aligned} \right\} \text{ at } T_2 = 312.685 \text{ K} \text{ or } 39.535^\circ\text{C}$$

$$\rightarrow Re = \frac{\rho V_7 d}{\mu}$$

$$Re = \frac{0.6967 \times 367.34 \times 1.4224}{10.4 \times 10^{-6}}$$

$$Re = 2.8383 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{2.8383 \times 10^7} \right] \right]^2}$$

$$f = 0.009936$$

$$\rightarrow h_{L \text{ major}} = 0.009936 \times \frac{5000}{1.4224} \times \frac{(367.34)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 240214.76 \text{ m}$$

→ Elbow El 9:

$$h_{L \text{ minor}} = 0.4 \times \frac{(367.34)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 2751.0811 \text{ m}$$

Spin Boldak (65 km):

$$\rightarrow Re = \frac{\rho_{20} V_7 d}{\mu_{20}}$$

$$Re = \frac{0.703 \times 367.34 \times 1.4224}{10.5 \times 10^{-6}}$$

$$Re = 3.4983 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.4983 \times 10^7} \right] \right]^2}$$

$$f = 0.009909$$

$$\rightarrow h_{L \text{ major}} = 0.009909 \times \frac{65000}{1.4224} \times \frac{(367.34)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 3114377.5744 \text{ m}$$

Wesh (65 km):

$$\rightarrow Re = \frac{\rho_{21} V_7 d}{\mu_{21}}$$

$$Re = \frac{0.706 \times 367.34 \times 1.4224}{10.46 \times 10^{-6}}$$

$$Re = 3.5267 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.5267 \times 10^7} \right] \right]^2}$$

$$f = 0.009908$$

$$\rightarrow h_{L \text{ major}} = 0.009908 \times \frac{65000}{1.4224} \times \frac{(367.34)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 3114084.5317 \text{ m}$$

Chaman: (Before 8% branch)

$\rightarrow$ Elbow El 10:

$$h_{L \text{ minor}} = 0.4 \times \frac{(367.34)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 2751.0811 \text{ m}$$

$\rightarrow$ Tee:

$$h_{L \text{ minor}} = 1 \times \frac{(367.34)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 6877.6 \text{ m}$$

$\rightarrow$ Solving Branch:

$$Q_8 = 583.72 \times \frac{8}{100} = 46.6976 \text{ m}^3/\text{s}$$

$$Q_9 = 583.72 - 46.6976 = 537.0224 \text{ m}^3/\text{s}$$

$$V_8 = \frac{Q_8}{A} = \frac{46.6976}{\frac{\pi}{4} (1.4224)^2} = 29.39 \text{ m/s}$$

$$V_9 = \frac{Q_9}{A} = \frac{537.0224}{\frac{\pi}{4} (1.4224)^2} = 337.955 \text{ m/s}$$

(After 8% branch)

$\rightarrow$ branch ball valve:

$$h_{L \text{ minor}} = 0.05 \frac{(29.39)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 2.2 \text{ m}$$

$$\rightarrow Re = \frac{\rho_{22} V_9 d}{\mu_{22}}$$

$$Re = \frac{0.6995 \times 337.955 \times 1.4224}{10.36 \times 10^{-3}}$$

$$Re = 3.2457 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[0 - 1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.2457 \times 10^7} \right] \right]^2}$$

$$f = 0.009918$$

$$\rightarrow h_{L \text{ major}} = 0.009918 \times \frac{50000}{1.4224} \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2029541.9572 \text{ m}$$

$\rightarrow$ Ball valve:

$$h_{L \text{ minor}} = 0.05 \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 291.0644 \text{ m}$$

$\rightarrow$ Elbow El II:

$$h_{L \text{ minor}} = 0.4 \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 2328.551 \text{ m}$$

Mastung (55 km):

$$\rightarrow Re = \frac{\rho_{23} V_{g d}}{\mu_{23}}$$

$$Re = \frac{0.7107 \times 337.955 \times 1.4224}{10.17 \times 10^{-6}}$$

$$Re = 3.3593 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.3593 \times 10^7} \right] \right]^2}$$

$$f = 0.009914$$

$$\rightarrow h_{L \text{ major}} = 0.009914 \times \frac{55000}{1.4224} \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2231547.8882 \text{ m}$$

Quetta:

$$\rightarrow Re = \frac{\rho_{24} V_{g d}}{\mu_{24}}$$

$$Re = \frac{0.7035 \times 337.955 \times 1.4224}{10.28 \times 10^{-6}}$$

$$Re = 3.2897 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.2897 \times 10^7} \right] \right]^2}$$

$$f = 0.009916$$

$$\rightarrow h_{L \text{ major}} = 0.009916 \times \frac{92000}{1.4224} \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 3733729.6844 \text{ m}$$

→ Elbow El 12:

$$h_{L \text{ minor}} = 0.4 \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 2328.5151 \text{ m}$$

→ Elbow El 13:

$$h_{L \text{ minor}} = 2328.5151 \text{ m (same as El 12)}$$

Sibi (56.25 km):

$$\rightarrow Re = \frac{\rho_{25} V_9 d}{\mu_{25}}$$

$$Re = \frac{0.6833 \times 337.955 \times 1.4224}{10.62 \times 10^{-5}}$$

$$Re = 3.0929 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{8.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.0929 \times 10^7} \right] \right]^2}$$

$$f = 0.009924$$

$$\rightarrow h_{L \text{ major}} = 2284645.9874 \text{ m}$$

Dera Bugti (56.25 km):

$$\rightarrow Re = \frac{\rho_{26} V_9 d}{\mu_{26}}$$

$$Re = \frac{0.6821 \times 337.955 \times 1.4224}{10.64 \times 10^{-6}}$$

$$Re = 3.0817 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.0817 \times 10^7} \right] \right]^2}$$

$$f = 0.009925$$

$$\rightarrow h_{L \text{ major}} = 0.009925 \times \frac{56250}{1.4224} \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2284755.1792 \text{ m}$$

Rajanpur (56.25 km):

$$Re = 3.0929 \times 10^7$$

$$f = 0.009924$$

$$h_{L \text{ major}} = 2284645.9874 \text{ m}$$

Same P , μ and d as sibi.

Dera Ghazi Khan (56.25 km):

$$\rightarrow Re = \frac{P_{28} V_q d}{\mu_{28}}$$

$$Re = \frac{0.6845 \times 337.955 \times 1.4224}{10.6 \times 10^{-6}}$$

$$Re = 3.1042 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.1042 \times 10^7} \right] \right]^2}$$

$$f = 0.009924$$

$$\rightarrow h_{L \text{ major}} = 0.009924 \times \frac{56250}{1.4224} \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2284537.1529 \text{ m}$$

Taunsa Sharif (56.25 km):

$$\rightarrow Re = \frac{P_{29} \times V_q \times d}{\mu_{29}}$$

$$Re = \frac{0.6858 \times 337.955 \times 1.4224}{10.58 \times 10^{-6}}$$

$$Re = 3.1160 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.1160 \times 10^7} \right] \right]^2}$$

$$f = 0.009923$$

$$\rightarrow h_{L \text{ major}} = 0.009923 \times \frac{56250}{1.4224} \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2284424.338 \text{ m}$$

Muzaffargarh (56.25 km)

and Kot Addy (56.25 km):

$$Re = 3.0929 \times 10^7$$

$$f = 0.009924$$

$$h_{L \text{ major}} = 2284645.9874 \text{ m}$$

Both have same ρ , μ and d as Sibi.

Multan:

$$\rightarrow Re = \frac{\rho_{32} \times V_a \times d}{\mu_{32}}$$

$$Re = \frac{0.6816 \times 337.9551 \times 1.4224}{10.65 \times 10^{-6}}$$

$$Re = 3.0765 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{3.0765 \times 10^7} \right] \right]^2}$$

$$f = 0.009925$$

$$\rightarrow h_{L \text{ major}} = 0.009925 \times \frac{56250}{1.4224} \times \frac{(337.995)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 2284805.4926 \text{ m}$$

Elbow El 14:

$$h_{L \text{ minor}} = 0.4 \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 2328.5151 \text{ m}$$

Tee:

$$h_{L \text{ minor}} = 1 \times \frac{(337.955)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 5821.2877 \text{ m}$$

Solving 35% branch:

$$Q_{10} = 537.0224 \times \frac{35}{100} = 187.95784 \text{ m}$$

$$Q_{11} = 537.0224 - 187.95784 = 349.06456 \text{ m}$$

$$V_a = \frac{Q_{10}}{A} = \frac{187.95784}{\frac{\pi}{4} (1.4224)^2} = 118.28 \text{ m/s}$$

$$V_{11} = \frac{Q_{11}}{A} = \frac{349.06456}{\pi/4 (1.4224)^2} = 219.6708 \text{ m/s}$$

→ Elbow El 15:

$$h_{L \text{ minor}} = 0.4 \times \frac{(219.6708)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 983.7976 \text{ m}$$

→ Ball Valve (branch):

$$h_{L \text{ minor}} = 0.05 \times \frac{(219.6708)^2}{2 \times 9.81}$$

$$h_{L \text{ minor}} = 35.65 \text{ m}$$

Lodhran (61 km):

$$\rightarrow Re = \frac{\rho_{33} V_{11} d}{\mu_{33}}$$

$$Re = \frac{0.6833 \times 219.6708 \times 1.4224}{10.62 \times 10^{-6}}$$

$$Re = 2.0104 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{2.0104 \times 10^7} \right] \right]^2}$$

$$f = 0.009993$$

$$\rightarrow h_{L \text{ major}} = 0.009993 \times \frac{61000}{1.4224} \times \frac{(219.6708)^2}{2 \times 9.81}$$

$$h_{L \text{ major}} = 1054038.8627 \text{ m}$$

Vehari (61 km):

$$Re = 2.0104 \times 10^7$$

$$f = 0.009993$$

$$h_{L \text{ major}} = 1054038.8627 \text{ m}$$

Same ρ , μ and d as Lodhran.

Fazilka 61km:

$$\rightarrow Re = \frac{\rho_{35} \times V_n \times d}{\mu_{35}}$$

$$Re = \frac{0.69 \times 219.6708 \times 1.4224}{10.7 \times 10^{-6}}$$

$$Re = 2.0149 \times 10^7$$

$$\rightarrow f = \frac{1}{\left[-1.8 \log \left[\left(\frac{0.000046}{3.7 \times 1.4224} \right)^{1.11} + \frac{6.9}{2.0149 \times 10^7} \right] \right]^2}$$

$$f = 0.0099931$$

$$\rightarrow h_{L \text{ major}} = 0.0099931 \times \frac{61000}{1.4224} \times \frac{(219.6708)^2}{2 \times 9.81}$$

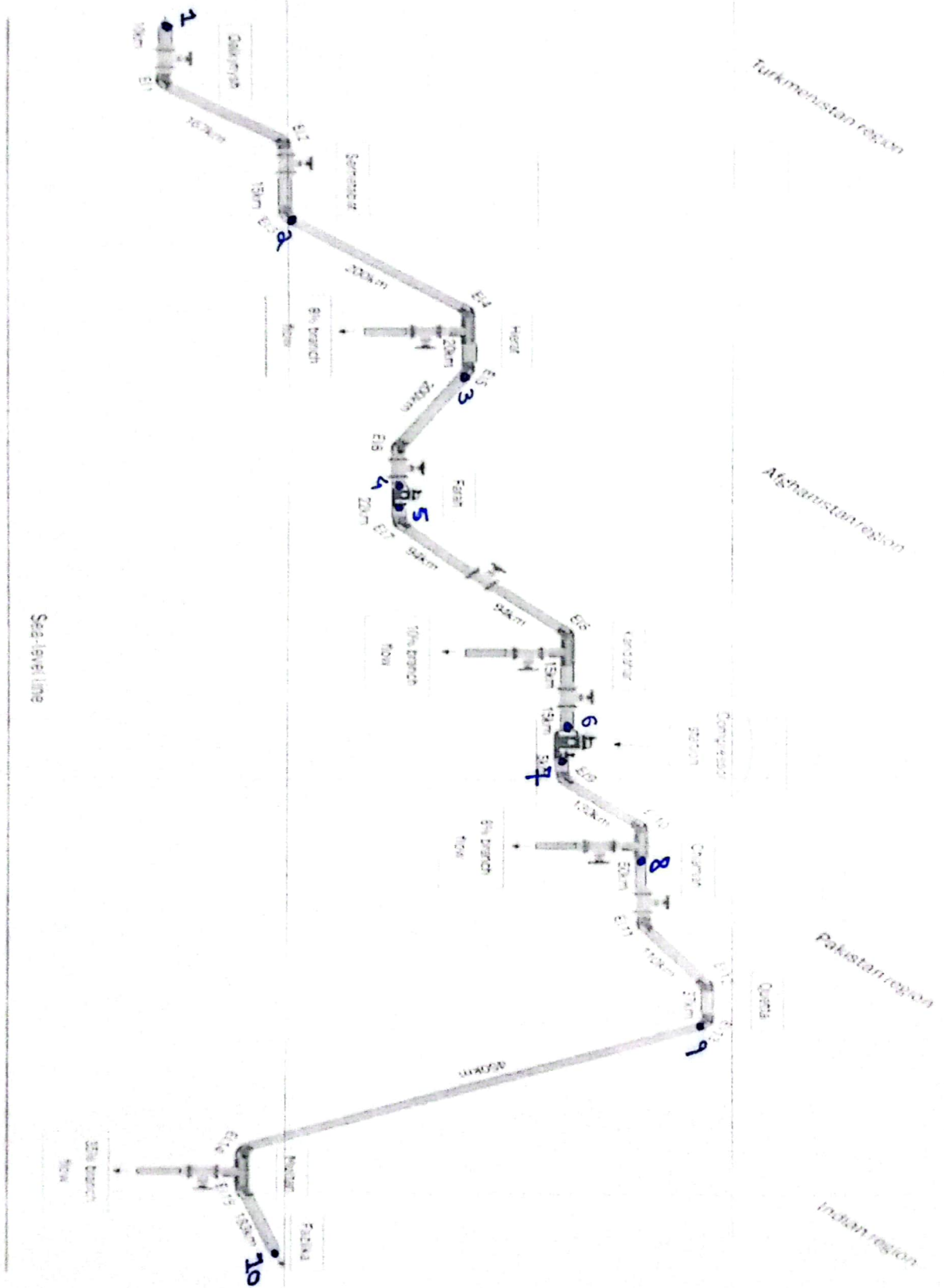
$$h_{L \text{ major}} = 1053992.9309 \text{ m}$$

For Pressure Drop, EGL and HGL

Fluid Mechanics

DMAE

BEME-IV-B



PRESSURE DROP

$$P_1 = 100 \text{ atm} = 10132500 \text{ Pa}$$

Point 1 and 2: (From Entrance to El 3)

$$P_1 + \frac{1}{2} \rho V_1^2 + \gamma_1 z_1 = P_2 + \frac{1}{2} \rho V_2^2 + \gamma_2 z_2 + \gamma h_L \quad \text{--- (i)}$$

$$V_1 = V_2$$

$$\gamma h_L = \gamma h_{L \text{ major}} + \gamma h_{L \text{ minor}}$$

$$\gamma h_{L \text{ major}} = \gamma_1 h_{L20} + \gamma_2 h_{L40} + \gamma_3 h_{L40} + \gamma_4 h_{L40} + \gamma_5 h_{L47} + \gamma_6 h_{L25}$$

$$\gamma h_{L \text{ major}} = 10418569.25 + 42134500.82 + 42015917.06 + 42075073.6 + 49508356.63 + 15215031.51$$

$$\gamma h_{L \text{ major}} = 201367448.9 \text{ Pa}$$

$$\gamma h_{L \text{ minor}} = \gamma_1 h_{L \text{ valve}} + \gamma_1 h_{L \text{ El1}} + \gamma_2 h_{L \text{ El2}} + \gamma_2 h_{L \text{ valve}} + \gamma_2 h_{L \text{ El3}}$$

$$\gamma h_{L \text{ minor}} = 7517.89268 + 60189.14745 + (2 \times 58541.31776) + 7312.071411$$

$$\gamma h_{L \text{ minor}} = 192101.7471$$

$$(i) \rightarrow P_1 - P_2 = \gamma_2 z_2 - \gamma_1 z_1 + \gamma h_{L \text{ major}} + \gamma h_{L \text{ minor}}$$

$$P_1 - P_2 = 4946.44725 - 1531.8315 + 201367448.9 + 192101.7471$$

$$P_1 - P_2 = 201562965.3 \text{ Pa} \quad \text{--- (1)}$$

$$P_2 = P_1 - 201562965.3$$

$$P_2 = 10132500 - 201562965.3$$

$$P_2 = -191430465.3 \text{ Pa}$$

Point 2 and 3: (After E13 to E15)

$$P_2 - P_3 = \gamma_3 z_3 - \gamma_2 z_2 + \frac{1}{2} [\rho_3 v_3^2 - \rho_2 v_2^2] + \gamma h_{L \text{ major}} + \gamma h_{L \text{ minor}}$$

$$\begin{aligned} \gamma h_{L \text{ major}} &= \gamma_7 h_{L_{66}} + \gamma_8 h_{L_{66}} + \gamma_9 h_L + \gamma_{10} h_{L_{20km}} \\ &= 70790369.31 + 71085871.29 + 72128254.12 + 17966049.18 \\ &= 231970544 \text{ Pa} \end{aligned}$$

$$\begin{aligned} h_{L \text{ minor}} &= \gamma_{10} h_{L_{E14}} + \gamma_{10} h_{L_{Tee}} + \gamma_{10} h_{L_{valve}} + \gamma_{10} h_{L_{E15}} \\ &= 61229.8898 + 153074.7046 + 49.019 + 51824.97204 \\ &= 266178.578 \text{ Pa} \end{aligned}$$

$$\begin{aligned} P_2 - P_3 &= 6371.7912 - 4946.4425 + \frac{1}{2} [(0.706)(605.8)^2 - (0.675)(658.5)^2] \\ &\quad + 231970544 + 266178.578 \end{aligned}$$

$$\boxed{P_2 - P_3 = 232221349.2} \quad \text{--- (2)}$$

$$P_3 = P_2 - 232221349.2$$

$$P_3 = -191430465.3 - 232221349.2$$

$$\boxed{P_3 = -423651814.5}$$

Point 3 and 4: (After E15 to compressor inlet)

$$P_3 - P_4 = \gamma_4 z_4 - \gamma_3 z_3 + \gamma h_{L \text{ major}} + \gamma h_{L \text{ minor}}$$

$$\begin{aligned} \gamma h_{L \text{ major}} &= \gamma_{11} h_{L_{50}} + \gamma_{12} h_{L_{50}} + \gamma_{13} h_{L_{50}} + \gamma_{14} h_{L_{50}} \\ &= 45417303.93 + 45921035.96 + 45166214.05 + \\ &\quad 44565870.77 \\ &= 181070425 \text{ Pa} \end{aligned}$$

$$\begin{aligned} \gamma h_{L \text{ minor}} &= \gamma_4 h_{L_{E16}} + \gamma_4 h_{L_{valve}} \\ &= 51237.72023 + 6404.714686 \\ &= 57642.4349 \text{ Pa} \end{aligned}$$

$$P_3 - P_4 = 4450.797 - 6391.7912 + 181070425 + 57642.4349$$

$$\boxed{P_3 - P_4 = 181126146.2 \text{ Pa}} \quad \text{--- (3)}$$

$$P_4 = P_3 - 181126146.2$$

$$\boxed{P_4 = -604777960.7 \text{ Pa}}$$

Compressor Station:

$$r_p = \frac{P_{\text{outlet}}}{P_4}$$

$$P_{\text{outlet}} = r_p \times |P_4|$$

$$P_{\text{outlet}} = 1.6 \times 604777960.7$$

$$\boxed{P_{\text{outlet}} = 967644737.1 \text{ Pa}}$$

$$P_5 = P_4 + P_{\text{outlet}}$$

$$P_5 = -604777960.7 + 967644737.1$$

$$\boxed{P_5 = 362866776.4 \text{ Pa}}$$

Point 5 and 6: (After compressor to inlet of 2nd Compressor)

$$P_5 - P_6 = \gamma_6 z_6 - \gamma_5 z_5 + \frac{1}{2} [P_6 v_6^2 - P_5 v_5^2] + h_{L_{\text{major}}} + h_{L_{\text{minor}}}$$

$$\begin{aligned} h_{L_{\text{major}}} &= \gamma_{15} h_{L_{22}} + \gamma_{16} h_{L_{63}} + \gamma_{17} h_{L_{63}} + \gamma_{18} h_{L_{62}} + \gamma_{19} h_{L_{30}} \\ &= 13115139.35 + 2 \times (37612534.24) + 36756108.77 + 14441369.34 \\ &= 139540686 \text{ Pa} \end{aligned}$$

$$\begin{aligned} h_{L_{\text{minor}}} &= \gamma_{15} h_{L_{E17}} + \gamma_{17} h_{L_{\text{valve}}} + \gamma_{19} h_{L_{E18}} + \gamma_{19} h_{L_{\text{Tee}}} + \gamma_{19} h_{L_{\text{valve}}} \\ &\quad + \gamma_{19} h_{L_{\text{b.valve}}} \\ &= 21415.83761 + 4298.508518 + 21375.95138 + \\ &\quad 885503.80622 + 42.75 + 3462.8649 \\ &= 136099.719 \text{ Pa} \end{aligned}$$

$$P_5 - P_6 = 6943.98 - 4450.797 + \frac{1}{2} \left[(0.698)^2 (445.89)^2 - (0.698)(445.43)^2 \right] + 139540686 + 136099.719$$

$$P_5 - P_6 = 139662874.8 \text{ Pa} \quad \text{--- (4)}$$

$$P_6 = P_5 - 139662874.8$$

$$P_6 = 362866776.4 - 139662874.8$$

$$P_6 = 223203901.6 \text{ Pa}$$

Compressor Station:

$$P_{\text{outlet}} = r \times P_6$$

$$P_{\text{outlet}} = 1.6 \times 223203901.6$$

$$P_{\text{outlet}} = 357126242.6$$

$$P_7 = P_6 + P_{\text{outlet}}$$

$$P_7 = 22320390.6 + 357126242.6$$

$$P_7 = 580330144.2 \text{ Pa}$$

Point a (7 and 8): [after compressor to Tee (chaman)]

$$P_7 - P_8 = \gamma_8 z_8 - \gamma_7 z_7 + \frac{1}{2} \left[\rho_8 v_8^2 - \rho_7 v_7^2 \right] + \gamma_{L\text{major}} + \gamma_{L\text{minor}}$$

$$\gamma_{L\text{major}} = \gamma_{19} h_{L5} + \gamma_{20} h_{L65} + \gamma_{24} h_{L65}$$

$$= 1641778.284 + 21478086.94 + 21567713.49$$

$$= 44687578.7 \text{ Pa}$$

$$h_{L\text{minor}} = \gamma_{19} h_{L119} + \gamma_{22} (h_{L110} + h_{L\text{Tee}} + h_{L\text{O. valve}})$$

$$= 18802.613 + 18878.17986 + 47194.745 + 15.0966$$

$$= 84890.6342 \text{ Pa}$$

$$P_7 - P_8 = 9119.724 - 6943.98 + \frac{1}{2} \left[(0.6995)^2 (337.95)^2 - (0.6967)^2 (445.89)^2 \right] + 44687578.7 + 84890.6342$$

$$P_7 - P_8 = 44772396 \text{ Pa} \quad \text{--- (5)}$$

$$P_8 = P_7 - 44772396$$

$$P_8 = 580330144.2 \text{ Pa} - 44772396 \text{ Pa}$$

$$P_8 = 535557748.2 \text{ Pa}$$

Point 8 and 9: (After Chaman Tee to El 13)

$$P_8 - P_9 = \gamma_9 z_9 - \gamma_8 z_8 + h_{L \text{ major}} + h_{L \text{ minor}}$$

$$\begin{aligned} \rightarrow h_{L \text{ major}} &= \gamma_{22} h_{L50} + \gamma_{23} h_{L55} + \gamma_{24} h_{L92} \\ &= 13926904.23 + 15558277.68 + 25767719.35 \\ &= 55252901.3 \text{ Pa} \end{aligned}$$

$$\begin{aligned} \rightarrow h_{L \text{ minor}} &= \gamma_{22} (h_{L \text{ Ball valve}} + h_{L \text{ El 11}}) + \gamma_{24} (h_{L \text{ El 12}} + h_{L \text{ El 13}}) \\ &= 1997.31 + 15978.49 + 2(16069.86) \\ &= 50115.5289 \text{ Pa} \end{aligned}$$

$$P_8 - P_9 = 11594.2428 - 9119.72 + 55252901.3 + 50115.5289$$

$$P_8 - P_9 = 55305491.3 \quad \text{--- (6)}$$

$$P_9 = P_8 - 55305491.3$$

$$P_9 = 535557748.2 - 55305491.3$$

$$P_9 = 480252256.9 \text{ Pa}$$

Point 9 and 10: (After El 13 to Fazilka)

$$P_9 - P_{10} = \gamma_{10} z_{10} - \gamma_9 z_9 + \frac{1}{2} [P_{10} V_{10}^2 - P_9 V_9^2] + h_{L \text{ major}} + h_{L \text{ minor}}$$

$$\begin{aligned} \rightarrow h_{L \text{ major}} &= \gamma_{25} h_{36.25} + \gamma_{26} h_{L56.25} + \gamma_{27} h_{L56.25} + \gamma_{28} h_{L56.25} \\ &\quad + \gamma_{29} h_{L56.25} + \gamma_{30} h_{L56.25} + \gamma_{31} h_{L56.25} + \gamma_{32} h_{L56.25} \\ &\quad + \gamma_{33} h_{L61} + \gamma_{34} h_{L61} + \gamma_{35} h_{L61} \end{aligned}$$

$$\begin{aligned}
 &= 4(15314377.3) + 15288213.09 + 15340541.33 + \\
 &15368917.04 + 15277342.79 + 2(7065404.845) + \\
 &7134372.75 \\
 &= 143797706 \text{ Pa}
 \end{aligned}$$

$$\begin{aligned}
 \delta h_{L \text{ minor}} &= \gamma_{32} (h_{L \text{ El4}} + h_{L \text{ Tee}} + h_{L \text{ p. valve}} + h_{L \text{ El5}}) \\
 &= 15569.6 + 38924.01 + 238.37 + 6578.16 \\
 &= 61310.1561
 \end{aligned}$$

$$\begin{aligned}
 P_9 - P_{10} &= 1231.9398 - 11594.2428 + \frac{1}{2} \left[(0.69)(219.6708)^2 - \right. \\
 &\quad \left. (0.7035)(337.955)^2 \right] + 143797706 + 61310.1561
 \end{aligned}$$

$$\boxed{P_9 - P_{10} = 143825127 \text{ Pa}}$$

$$P_{10} = P_9 - 143825127$$

$$P_{10} = 480252256.9 - 143825127$$

$$\boxed{P_{10} = 336427129.9 \text{ Pa}}$$

Hydraulic Grade Line

Point 1:

$$HGL_1 = \frac{P_1}{\gamma} + z$$

$$HGL_1 = 1488291.9 + 225$$

$$HGL_1 = 1488517 \text{ m}$$

Point 2:

$$HGL_2 = \frac{P_2}{\gamma} + z$$

$$HGL_2 = -28909346.5 + 747$$

$$HGL_2 = -28908599.5 \text{ m}$$

Point 3:

$$HGL_3 = \frac{P_3}{\gamma} + z$$

$$HGL_3 = -61169560.8 + 920$$

$$HGL_3 = -61168640.8 \text{ m}$$

Point 4:

$$HGL_4 = \frac{P_4}{\gamma} + z$$

$$HGL_4 = -88322535.1 + 650$$

$$HGL_4 = -88321885.1 \text{ m}$$

Point 5:

$$HGL_5 = \frac{P_5}{\gamma} + z$$

$$HGL_5 = 52993521.1 + 650$$

$$HGL_5 = 52994171.1 \text{ m}$$

Point 6:

$$HGL_6 = P_6/\gamma + z$$

$$HGL_6 = 32657802.9 + 1016$$

$$HGL_6 = 32658818.9m$$

Point 7:

$$HGL_7 = P_7/\gamma + z$$

$$HGL_7 = 84910287.6 + 1016$$

$$HGL_7 = 84911303.6m$$

Point 8:

$$HGL_8 = P_8/\gamma + z$$

$$HGL_8 = 78045807.9 + 1329$$

$$HGL_8 = 78047136.9m$$

Point 9:

$$HGL_9 = P_9/\gamma + z$$

$$HGL_9 = 69588312.5 + 1680$$

$$HGL_9 = 69589992.5m$$

Point 10:

$$HGL_{10} = P_{10}/\gamma + z$$

$$HGL_{10} = 49701891 + 182$$

$$HGL_{10} = 49702073m$$

Energy Grade Line

Point 1:

$$EGL_1 = HGL_1 + \frac{V_1^2}{2g}$$

$$EGL_1 = 1488516.96 + \frac{(658.5)^2}{2 \times 9.81}$$

$$EGL_1 = 1510617.989 \text{ m}$$

Point 2:

$$EGL_2 = HGL_2 + \frac{V_2^2}{2g}$$

$$EGL_2 = -28908599.5 + \frac{(658.5)^2}{2 \times 9.81}$$

$$EGL_2 = -28886498.49 \text{ m}$$

Point 3:

$$EGL_3 = HGL_3 + \frac{V_3^2}{2g}$$

$$EGL_3 = -61168640.8 + \frac{(605.8)^2}{2 \times 9.81}$$

$$EGL_3 = -61149935.74$$

Point 4:

$$EGL_4 = HGL_4 + \frac{V_4^2}{2g}$$

$$EGL_4 = -88321885.1 + \frac{(605.8)^2}{2 \times 9.81}$$

$$EGL_4 = -88303180.07 \text{ m}$$

Point 5:

$$EGL_5 = HGL_5 + \frac{V_5^2}{2g}$$

$$EGL_5 = 52994171.1 + \frac{(495.43)^2}{2 \times 9.81}$$

$$EGL_5 = 53006681.32 \text{ m}$$

Point 6:

$$EGL_6 = HGL_6 + \frac{V_6^2}{2g}$$

$$EGL_6 = 32658818.9 + \frac{(445.89)^2}{2 \times 9.81}$$

$$EGL_6 = 32668952.35 \text{ m}$$

Point 7:

$$EGL_7 = HGL_7 + \frac{V_7^2}{2g}$$

$$EGL_7 = 84911303.6 + \frac{(367.34)^2}{2 \times 9.81}$$

$$EGL_7 = 84918181.21 \text{ m}$$

Point 8:

$$EGL_8 = HGL_8 + \frac{V_8^2}{2g}$$

$$EGL_8 = 78047136.9 + \frac{(337.955)^2}{2 \times 9.81}$$

$$EGL_8 = 78052958.19 \text{ m}$$

Point 9:

$$EGL_9 = HGL_9 + \frac{V_9^2}{2g}$$

$$EGL_9 = HGL_9 + 69589992.5 + \frac{(337.955)^2}{2 \times 9.81}$$

$$EGL_9 = 69595813.82 \text{ m}$$

Point 10:

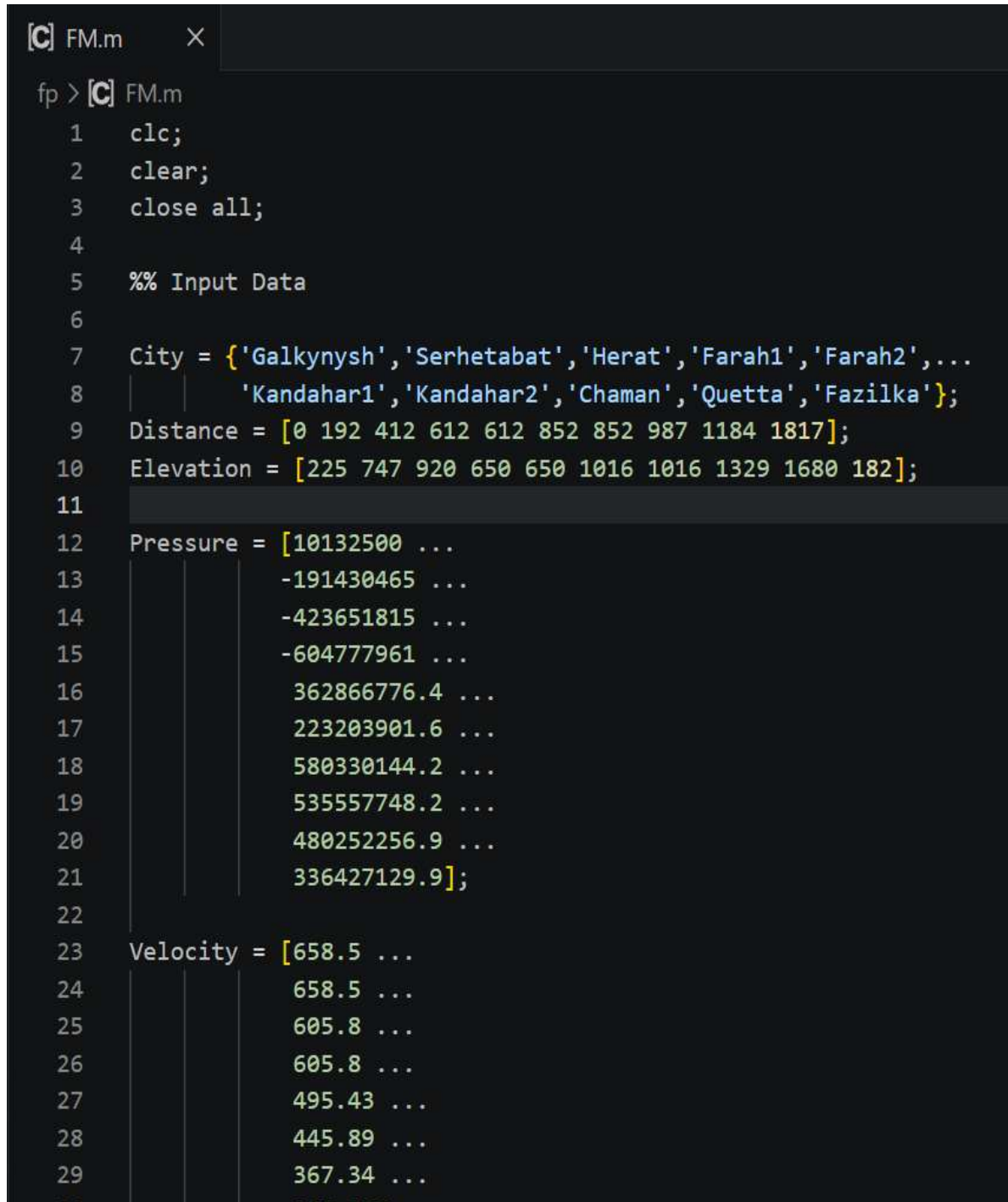
$$EGL_{10} = HGL_{10} + \frac{V_{10}^2}{2g}$$

$$EGL_{10} = 49702073 + \frac{(219.6708)^2}{2 \times 9.81}$$

$$EGL_{10} = 49704532.48 \text{ m}$$

MATLAB:

MATLAB Code:



The image shows a MATLAB script in a code editor window titled 'FM.m'. The script is as follows:

```
[C] FM.m ×
fp > [C] FM.m
1  clc;
2  clear;
3  close all;
4
5  %% Input Data
6
7  City = {'Galkynysh','Serhetabat','Herat','Farah1','Farah2',...
8         'Kandahar1','Kandahar2','Chaman','Quetta','Fazilka'};
9  Distance = [0 192 412 612 612 852 852 987 1184 1817];
10 Elevation = [225 747 920 650 650 1016 1016 1329 1680 182];
11
12 Pressure = [10132500 ...
13             -191430465 ...
14             -423651815 ...
15             -604777961 ...
16             362866776.4 ...
17             223203901.6 ...
18             580330144.2 ...
19             535557748.2 ...
20             480252256.9 ...
21             336427129.9];
22
23 Velocity = [658.5 ...
24             658.5 ...
25             605.8 ...
26             605.8 ...
27             495.43 ...
28             445.89 ...
29             367.34 ...
30             337.33 ...]
```

Figure 6

```
[C] FM.m  X
fp > [C] FM.m
23  Velocity = [658.5 ...
24             367.34 ...
29             337.955 ...
30             337.955 ...
31             219.6708];
32
33
34  %% Constants
35
36  gamma = 6.808;      % N/m^3 (Use your actual gamma value)
37  g = 9.81;          % m/s^2
38
39  %% Calculations
40
41  P_gamma = Pressure ./ gamma;
42
43  V_squared = Velocity.^2;
44
45  VelocityHead = V_squared ./ (2*g);
46
47  HGL = P_gamma + Elevation;
48
49  EGL = HGL + VelocityHead;
50
51  %% Pressure Drop Calculation
52
53  PressureDrop = zeros(length(Pressure)-1,1);
54
55  for i = 1:length(Pressure)-1
56      PressureDrop(i) = Pressure(i) - Pressure(i+1);
```

Figure 7


```
[C] FM.m X
fp > [C] FM.m
55   for i = 1:length(Pressure)-1
56
57   end
58
59   Distance_PD = Distance(2:end);
60
61   %% Display Results
62
63   Results = table( ...
64       Distance', ...
65       Elevation', ...
66       Pressure', ...
67       Velocity', ...
68       V_squared', ...
69       VelocityHead', ...
70       HGL', ...
71       EGL', ...
72       'VariableNames', ...
73       {'Distance_km','Elevation_m','Pressure_Pa','Velocity_ms',...
74        'V2','V2_2g','HGL','EGL'});
75
76   disp(Results);
77
78   %% Graph 1: HGL
79
80   figure;
81   plot(Distance,HGL,'-ob','LineWidth',2);
82   grid on;
83   xlabel('Distance (km)');
84   ylabel('HGL');
```

Figure 8

```
[C] FM.m ×
fp > [C] FM.m
84 ylabel('HGL');
85 title('Hydraulic Grade Line (HGL)');
86 set(gca,'FontSize',12);
87
88 %% Graph 2: EGL
89
90 figure;
91 plot(Distance,EGL,'-or','LineWidth',2);
92 grid on;
93 xlabel('Distance (km)');
94 ylabel('EGL');
95 title('Energy Grade Line (EGL)');
96 set(gca,'FontSize',12);
97
98 %% Graph 3: HGL and EGL Together
99
100 figure;
101 plot(Distance,HGL,'-ob','LineWidth',2);
102 hold on;
103 plot(Distance,EGL,'-sr','LineWidth',2);
104 grid on;
105 xlabel('Distance (km)');
106 ylabel('Head');
107 title('HGL and EGL');
108 legend('HGL','EGL','Location','best');
109 set(gca,'FontSize',12);
110
111 %% Graph 4: Pressure Drop
112
```

Figure 9

```

[C] FM.m  X
fp > [C] FM.m
112
113     figure;
114     plot(Distance_PD,PressureDrop,'-dk','LineWidth',2);
115     grid on;
116     xlabel('Distance (km)');
117     ylabel('Pressure Drop (Pa)');
118     title('Pressure Drop Along Pipeline');
119     set(gca,'FontSize',12);
120
121     %% Save Results to Excel
122
123     writetable(Results,'Pipeline_Results.xlsx');
124
125     disp('Results exported to Pipeline_Results.xlsx');

```

Figure 10

MATLAB Results:

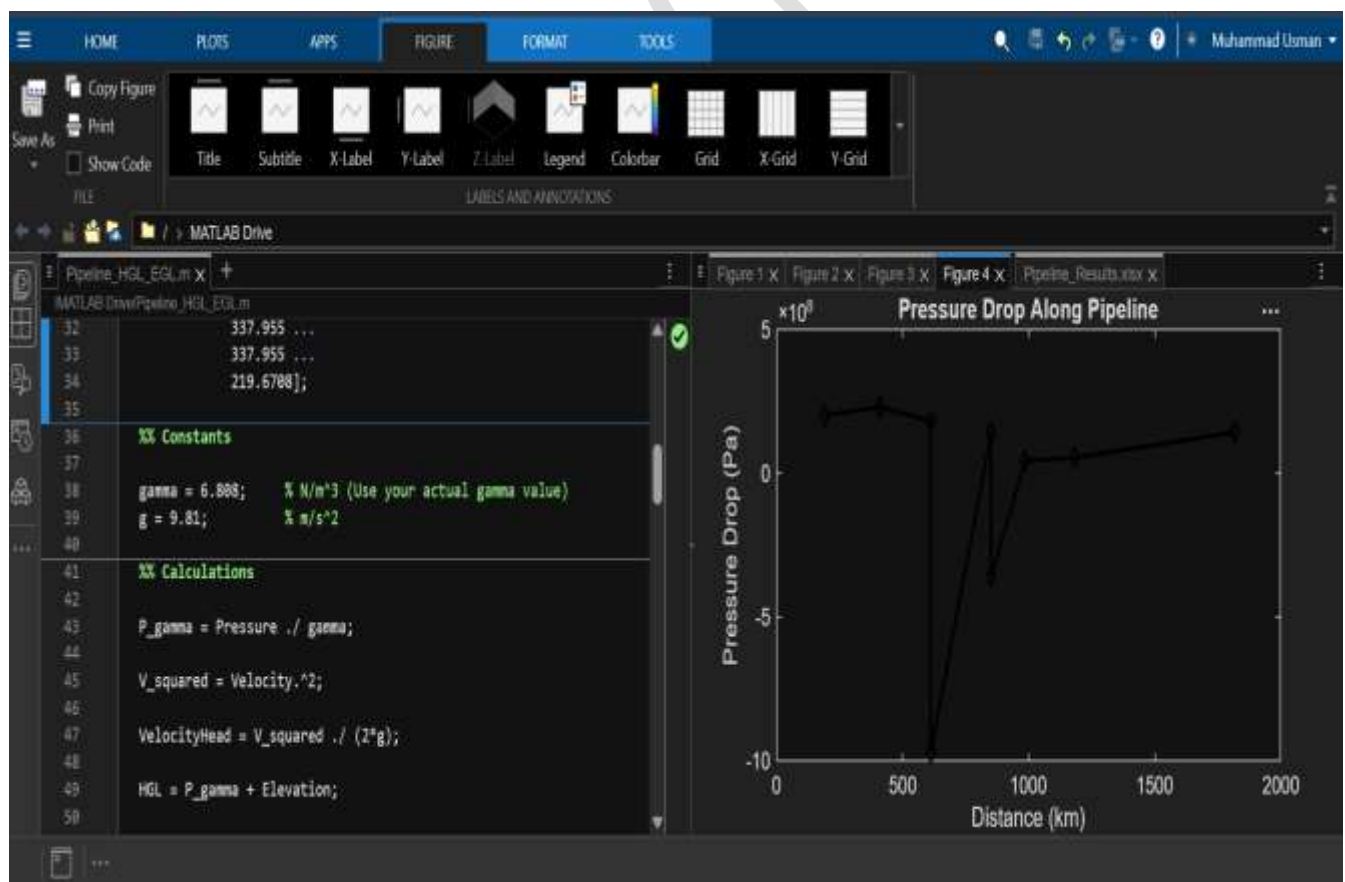


Figure 11

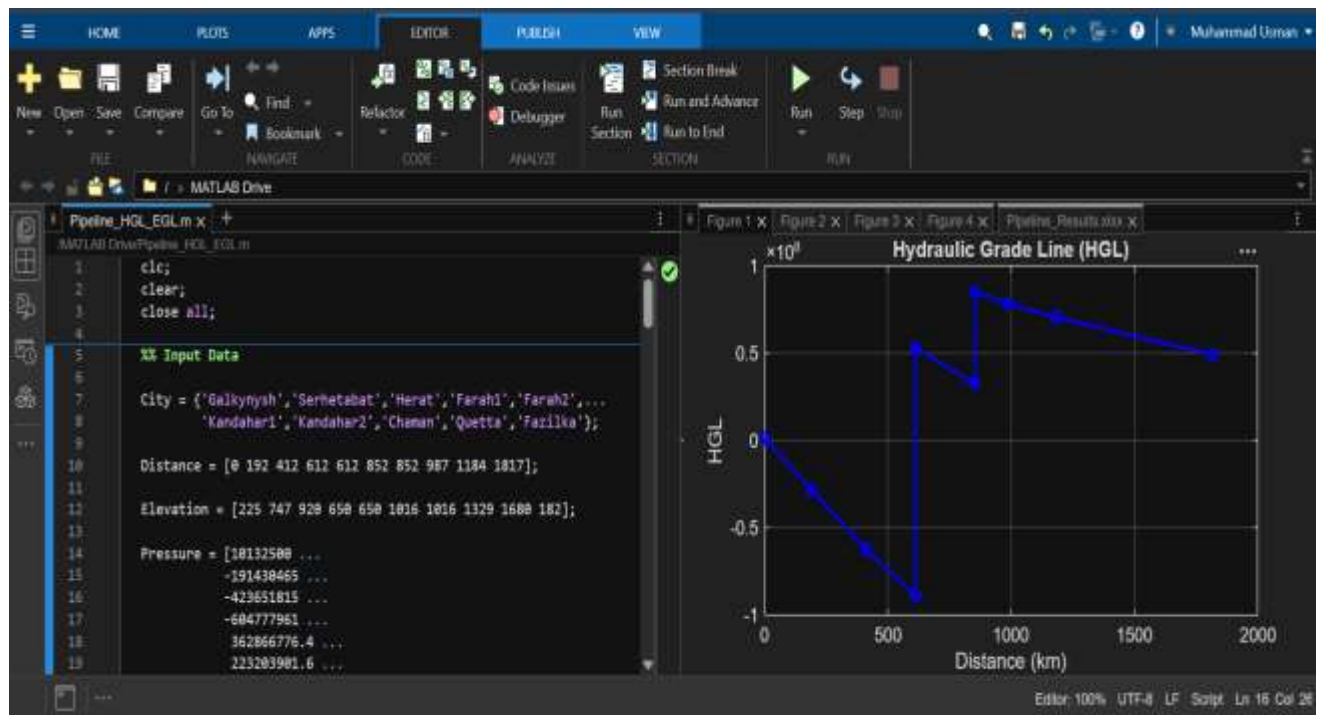


Figure 12

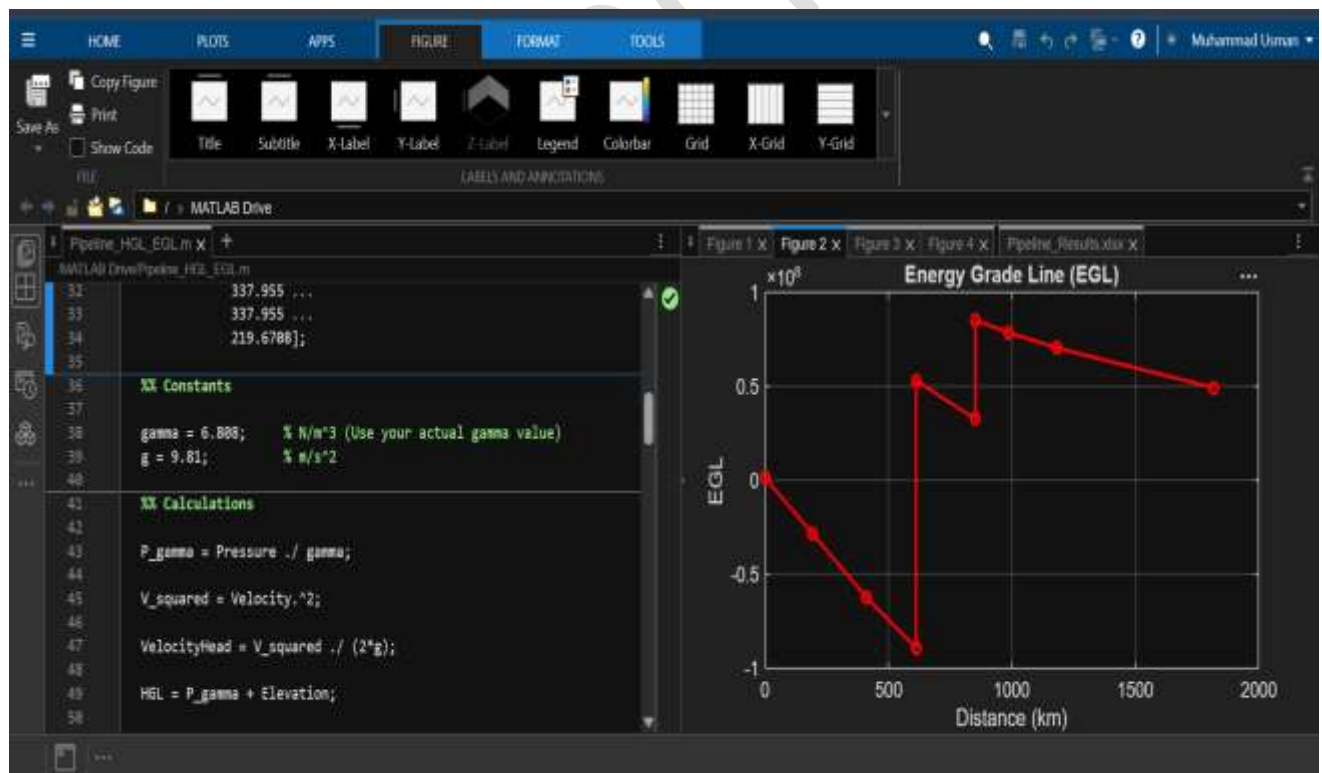


Figure 13

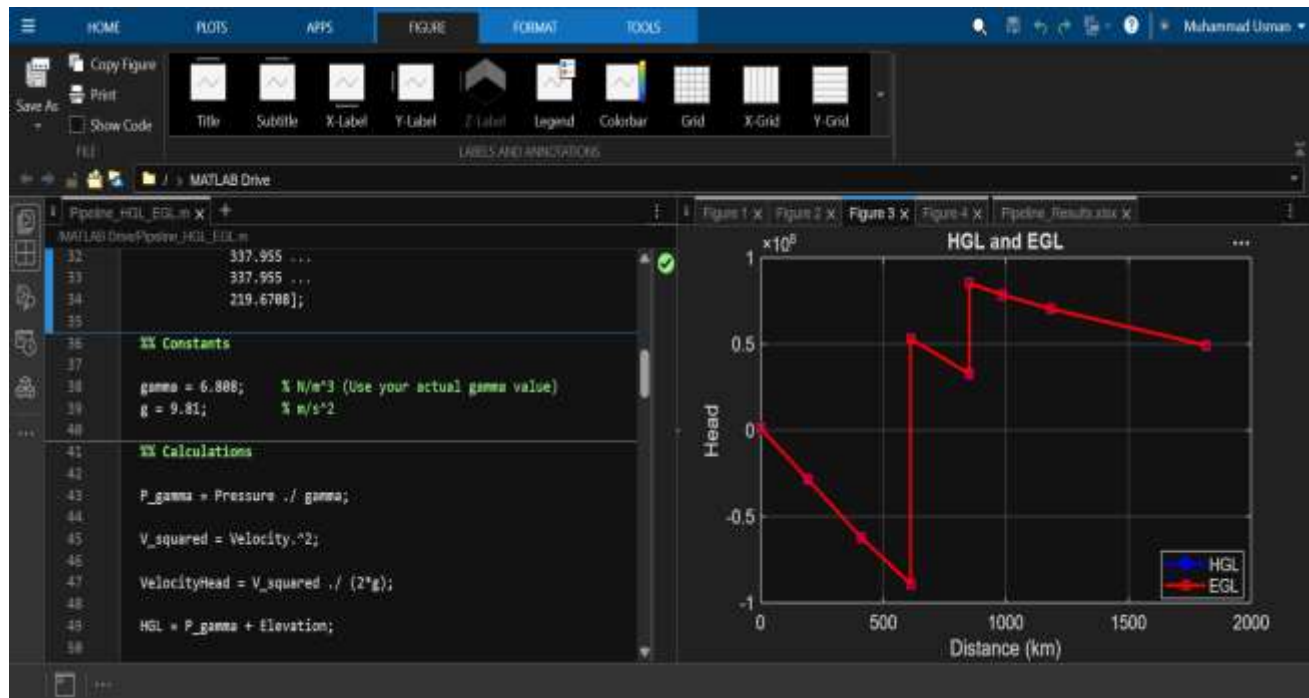


Figure 14



Figure 15

Discussion of Result:

- The results show that pressure decreases continuously along the pipeline due to frictional losses and changes in elevation.
- Larger pressure drops were observed in longer pipeline sections because friction losses increase with distance.
- The HGL graph showed the variation of pressure head throughout the pipeline and demonstrated the effect of energy losses during gas transportation.
- The EGL remained above the HGL at all locations because it includes both pressure energy and velocity energy of the flowing gas.
- Sudden increases in pressure, HGL, and EGL were observed at compressor station locations, indicating the restoration of energy lost during transmission.
- Elevation changes along the pipeline significantly influenced pressure behavior. Higher elevations required additional energy to maintain gas flow.
- The calculated pressure trends were consistent with fluid mechanics principles and the Darcy–Weisbach equation used in the analysis.
- The results highlighted the importance of compressor stations in maintaining adequate pressure levels over long transmission distances.
- The analysis demonstrated that both friction losses and terrain elevation must be considered when designing and operating long-distance natural gas pipelines.
- Overall, the calculated results provided a realistic representation of the hydraulic performance of the TAPI gas pipeline and achieved the objectives of the project.

Sources of Errors

- Compressor inlet temperature data was not available. Therefore, the average January temperature of the respective city was used to estimate natural gas density. This assumption may have introduced minor errors in density estimation and pressure calculations.
- Complete information about compressor stations along the TAPI pipeline was not available. Four compressor stations were not included in the analysis, which may have caused differences between the calculated and actual pressure values.
- The specifications, efficiency, and operating conditions of the compressors were unknown. As a result, assumptions had to be made regarding compressor performance, which may have affected the accuracy of the results.
- The exact pipeline route between all cities was not available. Some intermediate cities and pipeline segments may exist within certain sections of the pipeline, but these were neglected because the pipeline had to be divided into sections based on the available information.
- Methane was assumed to represent the natural gas flowing through the pipeline, and its properties were considered constant within each section. In reality, natural gas composition and properties can vary with operating conditions.
- Minor losses due to elbows, tees, valves, and other fittings were calculated using standard loss coefficients. Actual field conditions may result in different loss values.

- Pipe roughness was assumed using standard values for high-grade steel pipelines. Actual roughness may vary because of manufacturing tolerances, corrosion, deposits, and aging effects.
- The analysis was carried out under steady-flow conditions. In practice, gas pipelines may experience fluctuations in flow rate, pressure, and temperature during operation.
- Elevation data for different cities were obtained from publicly available sources and may not exactly represent the actual elevation profile of the pipeline route.
- Compressibility factor (Z-factor) values were estimated using standard correlations and charts. These methods provide approximate values and may introduce small calculation errors.
- MATLAB calculations were based on theoretical fluid mechanics equations and engineering assumptions. Therefore, slight deviations from actual industrial operating conditions are expected.

Applications:

Industrial Application:

1. Energy Production Industry

- The TAPI gas pipeline supplies natural gas for power generation plants.
- Helps industries maintain a stable energy supply for electricity production.
- Reduces dependence on expensive fuel alternatives.

2. Fertilizer Industry

- Natural gas transported through pipelines is used as a raw material in fertilizer manufacturing, especially ammonia and urea production.
- Supports agricultural productivity through fertilizer supply.

3. Manufacturing Industry

- Industries such as cement, steel, glass, and textiles use natural gas as a fuel source in manufacturing processes.
- Provides a cost-effective and cleaner energy option.

4. Oil and Gas Industry

- Pipeline systems are widely used for transporting oil and natural gas over long distances.
- Helps improve fuel distribution efficiency and energy accessibility.

Practical Applications

- Long-Distance Gas Transportation
- Pressure Monitoring and Flow Management
- Compressor Station Planning
- Pipeline Safety and Maintenance

Applications in Mechanical Engineering:

1. Fluid Mechanics

Helps mechanical engineers understand **fluid flow, pressure drop, and friction losses** in pipe systems. Applied in pipeline system design and analysis.

2. Pipeline Design Engineering

Used in selecting appropriate pipe diameter, material, and pressure requirements for safe gas transportation.

3. Hydraulic and Thermal Systems

Similar principles are applied in **HVAC systems, hydraulic machinery, boilers, pumps, and heat exchangers**.

4. Industrial System Optimization

Mechanical engineers use HGL and EGL concepts to improve system efficiency and reduce energy losses in fluid transport systems.

Recommendations:

1. Regular Pipeline Monitoring

- Continuous monitoring of pressure drops and flow conditions should be carried out to improve pipeline performance.
- Advanced monitoring systems can help detect pressure losses and leakage risks.

2. Proper Compressor Station Management

- Compressor stations should be strategically placed and regularly maintained to ensure stable gas flow.
- Efficient compressor operation can reduce energy losses in long-distance transportation.

3. Use of Advanced Hydraulic Analysis

- Modern hydraulic modeling techniques should be used for accurate prediction of pressure and energy variations.
- This can improve operational efficiency and reduce transportation losses.

4. Regular Maintenance and Inspection

- Pipeline systems should undergo regular inspection to reduce corrosion, leakage, and operational failures.

- Proper maintenance improves system reliability and safety.

5. Improved Route Planning

- Terrain elevation and environmental conditions should be carefully considered during pipeline planning to reduce unnecessary pressure losses.

Conclusion:

This project provided a theoretical understanding of the pressure drop, Hydraulic Grade Line (HGL), and Energy Grade Line (EGL) along the TAPI gas pipeline using fluid mechanics principles. The study highlighted that frictional resistance, terrain elevation, and pipeline length significantly influence gas flow and pressure variation in long-distance transmission systems. Theoretical analysis showed the importance of major and minor losses, as well as the role of compressor stations in maintaining continuous gas transportation. Overall, the project demonstrated how fluid mechanics concepts are practically applied in real-world engineering systems to improve the efficiency and reliability of large-scale gas pipelines.

References:

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- Weather of Pakistan : (<https://world-weather.info/forecast/pakistan/multan/month/>)
- Pipe roughness: (<https://moodychartcalc.com/blog/pipe-roughness-guide/>)
- Temperature Sources: (<https://www.worlddata.info/asia/turkmenistan/climate.php>)

Work Distribution:

Sr. No	Group Members Names	Assigned Work
1	Muhammad Usman Khan (241202)	Fluid Properties, Kandahar to Quetta, 1 Compressor Station
2	Areeba (241253)	Altitudes, K_L Values, Galkynysh to Herat, Description, Summary
3	Alishba (241212)	Herat to Kandahar, 1 Compressor Station
4	Bakhtawar Khan (231013)	Quetta to Fazilka, MATLAB (Energy and Hydraulic grade lines), Title Selection